AGMA 918–A93

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AMERICAN GEAR MANUFACTURERS ASSOCIATION

A Summary of Numerical Examples Demonstrating the Procedures for Calculating Geometry Factors for Spur and Helical Gears

AGMA INFORMATION SHEET

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918–A93, A Summary of Numerical Examples Demonstrating the Procedures for Calculating Geometry Factors for Spur and Helical Gears

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ABSTRACT

This information sheet provides numerical examples for calculating the pitting resistance geometry factor, I, and bending strength geometry factor, J, for typical gearsets that are generated by rack–type tools (hobs, rack cutters or generating grinding wheels) or pinion–type tools (disk–type shaper cutters). The numerical examples are shown in tabular form and provide the values for all variables as calculated using the procedures and equations in AGMA 908–B89. A flow chart, intended to assist in the development of a computer program for these variables, is also included.

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FOREWORD

[The foreword, footnotes, and annexes, if any, in this document are provided for informational purposes only and are not to be construed to be part of AGMA 918–A93, *A Summary of Numerical Examples Demonstrating the Procedures for Calculating Geometry Factors for Spur and Helical Gears.*]

This AGMA information sheet and related publications are based on typical or average data, conditions, or application.

This information sheet, AGMA 918–A93, was prepared to assist designers in the proper use and interpretation of AGMA 908–B89 and to assist in the development of computer programs when calculating geometry factors for pitting resistance, I, and bending strength, J. A flow chart provides a step by step procedure for the calculation of these factors, either manually or by computer program. Several examples are provided to demonstrate the calculation procedure for the various characteristics of geometry as described in AGMA 908–B89.

These include accurate and inaccurate spur gears, conventional and LACR helical gears, internal and external gears, double helical and herringbone (Sykes) gears, and addendum modifications. The calculation of J–factor for internal gears is not defined in AGMA 908–B89 and, therefore, is not covered in this information sheet. A tabulation of all calculated variables is provided for each example based on its design criteria. This provides the designer with known results to check against when calculating or programming these factors.

Suggestions for the improvement of this information sheet will be welcome. They should be sent to the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, Virginia, 22314.

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A Summary of Numerical Examples Demonstrating the Procedures for Calculating Geometry Factors for Spur and Helical Gears

1 Scope

This information sheet provides a set of numerical examples which calculate the geometry factor for pitting resistance, I, and bending strength, J, for a variety of gearsets selected to demonstrate the various gear geometries analyzed in AGMA 908–B89, *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical and Herringbone Gear Teeth.* A flow chart is also included to formalize the calculation procedures for the numerical examples and to assist in the practical application of AGMA 908–B89.

1.1 Numerical examples

Numerical examples were selected to demonstrate the following conditions: accurate and inaccurate spur gears, conventional and LACR (low axial contact ratio) helical gears, internal and external gears, double helical and herringbone (Sykes) gears and addendum modification. For simplification purposes and for demonstrating the effect on resulting geometry factors, similar examples were selected with different load locations (example 3.1.1 vs 3.1.2) and face widths (example 3.1.3 vs 3.1.4)

The results are presented in tabular form by providing the numerical results for each equation as presented in AGMA 908–B89 and appropriate to that gear geometry. Gear cutter data is presented for each component in each numerical example. All gearsets are functional and do not violate any of the exceptions stated in the scope of AGMA 908–B89. The examples used are for demonstration purposes only and are not intended to be recommendations for gearset design.

1.2 Flow chart

The flow chart provides a step by step procedure for calculating geometry factors, I and J, using the equations and instructions from AGMA 908–B89. The numerical value tables are formatted to coincide with the flow chart procedures.

1.3 Exceptions

A procedure for the calculation of bending strength geometry factor, J, for internal gears has not been established by AGMA. For this reason, numerical examples and flow chart procedures for such a calculation are not included.

2 Definitions and symbols

2.1 Definitions

The terms used, wherever applicable, conform to the following standards:

ANSI Y10.3–1968, *Letter Symbols for Quantities Used in Mechanics of Solids;*

AGMA 904–B89, *Metric Usage;*

AGMA 1012–F90, *Gear Nomenclature, Definitions of Terms with, Symbols.*

2. 2 Symbols

The symbols used in the geometry factor formulas are shown in table 1.

NOTE – The symbols, definitions and terminology used in this information sheet may differ from other AGMA documents. The user should not assume that familiar symbols can be used without a careful study of these definitions.

Units of measure are not shown in table 1 because the equations are in terms of unity normal module or unity normal diametral pitch.

Table 1 – Symbols used in equations

Symbols	Terms
$C_1, C_2, \ldots C_6$	distances along line of action (See figure 3–1 of AGMA 908–B89)
C_{n1}, C_{n4}, C_{n6}	distances along line of action of virtual spur gear
C_h	helical factor
C_r	operating center distance
C_ψ	helical overlap factor
D_i	inside diameter of internal gear
d	pinion operating pitch diameter
F	effective face width
F_1	gear type code
F_2	spur gear load sharing code
H	parameter for stress correction factor
h_{ao}	nominal tool addendum
h_F	height of Lewis parabola
I	pitting resistance geometry factor
J	bending strength geometry factor
K_f	stress correction factor
K_ψ	helix angle factor
L	parameter for stress correction factor
$L_{\min}$	minimum length of contact lines
M	parameter for stress correction factor
m_F	axial contact ratio
m_G	gear ratio
m_N	load sharing ratio
m_n	normal module
m_p	transverse contact ratio
n	virtual tooth number
n_o	virtual tooth number of tool
n_1, n_2	tooth number, pinion and gear
n_a	fractional part of m_F
n_c	tool tooth number
n_r	fractional part of m_p
P_{nd}	normal diametral pitch
p_b	transverse base pitch
p_N	normal base pitch
p_x	axial pitch
R_1, R_2	standard pitch radii, pinion and gear
R_{b1}, R_{b2}	base radii, pinion and gear
R_{bc}	base radius of tool

continued

Table 1 *(continued)*

Symbols	Terms
R_c	standard pitch radius of tool
R_{m1}	mean radius of pinion
R_{o1}, R_{o2}	addendum radii, pinion and gear, internal and external
R_{oc}	outside radius of tool
r_n, r_{n2}	reference pitch radii of virtual spur gear
r''_n	generating pitch radius of virtual spur gear
r_{no}	reference pitch radius of virtual tool
r''_{no}	generating pitch radius of virtual tool
r^s_{no}	radius to center "S" of tool tip radius
r_{na}, r_{na2}	virtual outside radii
r_{nb}, r_{nb2}	virtual base radii
r_{nbo}	virtual base radii of tool
r_{nL}	virtual load radius
s_F	tooth thickness at critical section
s_n	reference normal circular tooth thickness
s_{n1}, s_{n2}	reference normal circular tooth thickness, pinion and gear
s_{na}	tooth thickness at outside diameter
s_{no}	reference normal circular tooth thickness of tool
s_{ns}	standard tooth thickness, thinned for backlash
T_1	temporary variable
T_{o1}	temporary variable
u_s	stock allowance per side of tooth, for finishing
x	addendum modification coefficient at zero backlash
x_1, x_2	addendum modification coefficient, pinion and gear
x_g	generating rack shift coefficient
x_o	addendum modification coefficient of tool
x_{g1}, x_{g2}	generating rack shift coefficient, pinion and gear
Y	tooth form factor
y	iteration function
y'	derivative of iteration function
Z	active length of line of action
α_n	angle of surface, normal
α_{n1}	iteration angle
β_n	angle between tangent to fillet and tooth center line
Δs_n	amount gear tooth is thinned for backlash
δ_o	amount of protuberance, tool
δ_{ao}	amount of effective protuberance, tool
η_{nF}	ordinate of critical point "F"

continued

Table 1 *(concluded)*

Symbols	Terms
θ_n	angular displacement of gear
θ_{no}	angular displacement of tool
κ_F, κ_S	distance from pitch point to points "F" and "S"
λ_{ns}	angle to center "S" of tool tip radius
μ_{no}	auxiliary angle locating point "S"
ξ_{nF}	abscissa of critical point "F"
ρ_1, ρ_2	radii of curvature of profiles at point of contact stress calculation
ρ_{m1}, ρ_{m2}	radii of curvature of profile at mean radius
ρ_{ao}	tool tip radius
ρ_F	minimum radius of curvature of fillet curve
ϕ	standard transverse pressure angle
ϕ_n	standard normal pressure angle
ϕ''_n	generating pressure angle
ϕ''_{ni}	iteration value for generating pressure angle
ϕ_{nL}	load angle
ϕ_{npo}	pressure angle at radius where tool tooth is pointed
ϕ_{nr}	operating normal pressure angle
ϕ_{ns}	pressure angle at point "S" on tool
ϕ_{nW}	pressure angle at load application point
ϕ_r	operating transverse pressure angle
ψ	standard helix angle
ψ_b	base helix angle
ψ_r	operating helix angle
ω	angle of inclination of helical contact line

Subscripts			
o	tool	*n*	normal or virtual spur gear
1	pinion	*r*	operating or running
2	gear	–	absence of a subscript indicates transverse

3 Numerical examples

Eleven numerical examples, based on actual gearsets, are presented to demonstrate the calculation of both geometry factors, *I* and *J*, using the procedures outlined in AGMA 908–B89.

3.1 Examples

The following examples were selected to illustrate the various types of gearing and geometry features found in most of today's gearing.

3.1.1 Accurate spur gears

This example demonstrates a spur gearset which meets the criteria of table 5–1 in AGMA 908–B89 for load sharing and is therefore considered loaded at the highest point of single tooth contact.

3.1.2 Inaccurate spur gears

This example, which uses the same geometry as 3.1.1, does not meet the criteria in table 5–1 of AGMA 908–B89 for load sharing and is therefore considered to be loaded at the tip of the teeth.

3.1.3 Conventional helical gears

This example demonstrates a conventional helical gearset where the mesh face width is greater than the axial pitch. It also includes an addendum modification coefficient in the pinion and protuberance in the rack cutter used to generate both components.

3.1.4 Low axial contact ratio (LACR) helical gears

This example, which uses the same basic geometry as 3.1.3, demonstrates an LACR helical gearset where the face width has been reduced to less than the axial pitch. The effect on both geometry factors under these conditions can readily be seen.

3.1.5 Conventional helical gears, different tools

This example demonstrates a conventional helical gearset with addendum modification. The pinion is generated by a hob and the gear by a pinion type shaper cutter. Both cutters have protuberance.

3.1.6 Spur sun and planet gear

This example combines with 3.1.7 to demonstrate the geometry factor calculation for a spur sun/planet gear combination. Here, the I factor for the sun/planet mesh and the J factor for each component are calculated.

3.1.7 Spur planet and ring gear

This example combines with 3.1.6 and demonstrates the effect on the I factor when the same planet meshes with the internal ring gear of the same set. The calculated J factor for the planet in the planet/ring mesh is different from that in the sun/planet mesh (3.1.6). The J factor calculation for the ring gear is beyond the scope of this information sheet (see 1.3).

3.1.8 Helical sun and planet gear

This example combines with 3.1.9 to demonstrate the geometry factor calculation for a helical sun/planet gear combination. The I factor for the sun/planet mesh along with the J factor for each component is calculated.

3.1.9 Helical planet and ring gear

This example combines with 3.1.8 and demonstrates the effect on the I factor when the same planet meshes with the internal ring gear of the same set. The calculated J factor for the planet in the planet/ring mesh is different from that in the sun/planet mesh (3.1.8). The J factor calculation for the ring gear is beyond the scope of this information sheet (see 1.3).

3.1.10 Conventional double helical gears

This example demonstrates the method for considering the double face width encountered in this type of gearing.

3.1.11 Herringbone gears

This example demonstrates how transverse diametral pitch, usually associated with this type of gearing, is accommodated.

3.2 Tabulation of examples

Tables 2B through 12B tabulate all the information relating to each example as described in 3.1. The format is based on the flow chart as presented in clause 4 and includes all basic geometry (input data) and results of the calculations for every variable applicable to that gearset. These example were calculated to 14 significant digits and the results rounded as shown. For those variables found by iteration, the final iterative value is listed. See tables 2A through 12A for the specific value of each variable at each iteration loop. Figures 2 through 12 illustrate the various cutting tool profiles. Specific data relating to each gearset or component is listed in the individual example tables.

Table 2A – Accurate spur gears , example 3.1.1

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ_n''	0.014910	0.014910	0.014910	–	–	–
ϕ_{ni}''	0.358675	0.349386	0.349112	–	–	–
$\phi_{n(i+1)}''$	0.349386	0.349112	0.349111	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.630626	0.605889	0.605442	–	–
μ_{no}	0.000187	0.000257	0.000270	0.000271	–	–
κ_s	–1.325365	–1.589244	–1.645464	–1.646525	–	–
κ_F	–1.761865	–2.025744	–2.081964	–2.083025	–	–
θ_n	0.099917	0.113499	0.116204	0.116255	–	–
β_n	0.685481	0.517127	0.489685	0.489187	–	–
ξ_{nF}	1.179814	1.127191	1.119292	1.119150	–	–
η_{nF}	24.257896	24.334854	24.349198	24.349464	–	–
h_F	1.142721	1.065763	1.051418	1.051153	–	–
y'	4.452621	3.441053	3.322607	3.320554	–	–
y	0.689143	0.085121	0.001485	0.000000	–	–
$\|y\|$	0.689143	0.085121	0.001485	0.000000	–	–
α_{n1}	0.630626	0.605889	0.605442	0.605442	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ_n''	0.014894	0.014894	0.014894	–	–	–
ϕ_{ni}''	0.358546	0.349263	0.348989	–	–	–
$\phi_{n(i+1)}''$	0.349263	0.348989	0.348988	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.586171	0.541381	0.539939	0.539938	–
μ_{no}	0.000232	0.000349	0.000386	0.000387	0.000387	–
κ_s	–1.641409	–2.097854	–2.251830	–2.257242	–2.257248	–
κ_F	–2.077909	–2.534354	–2.688330	–2.693742	–2.693748	–
θ_n	0.053295	0.064581	0.068083	0.068205	0.068205	–
β_n	0.732103	0.521590	0.473298	0.471734	0.471733	–
ξ_{nF}	1.224450	1.158420	1.144696	1.144259	1.144259	–
η_{nF}	50.535758	50.627373	50.652661	50.653515	50.653516	–
h_F	1.239255	1.147640	1.122352	1.121498	1.121497	–
y'	5.034538	3.586125	3.374076	3.367760	3.367754	–
y	1.003017	0.160620	0.004867	0.000005	0.000000	–
$\|y\|$	1.003017	0.160620	0.004867	0.000005	0.000000	–
α_{n1}	0.586171	0.541381	0.539939	0.539938	0.539938	–

Table 2B – Accurate spur gears , example 3.1.1

Input data

Gearset

m_n	=	0.200000
ϕ_n	=	20.0000
ψ	=	0.0000
C_r	=	15.5000
F	=	2.5000
F_1	=	1
F_2	=	1

Pinion

n_1	=	51
n_{c1}	=	10000
R_{o1}	=	5.3225
x_1	=	0.1127
x_{o1}	=	0.0000

Cutter figure 2

h_{ao1}	=	1.4565
ρ_{ao1}	=	0.4365
δ_{ao1}	=	0.0099
Δs_{n1}	=	0.0215

Gear

n_2	=	104
n_{c2}	=	10000
R_{o2}	=	10.5774
x_2	=	–0.1127
x_{o2}	=	0.0000

Cutter figure 2

h_{ao2}	=	1.4565
ρ_{ao2}	=	0.4365
δ_{ao2}	=	0.0099
Δs_{n2}	=	0.0215

G2

ϕ_n	=	0.349066
ψ	=	0.000000
C_r	=	77.500000
F	=	12.500000
R_{o1}	=	26.612500
R_{o2}	=	52.887000
m_G	=	2.039216
R_1	=	25.500000
R_2	=	52.000000
ϕ	=	0.349066
R_{b1}	=	23.962162
R_{b2}	=	48.864016
ϕ_r	=	0.349066
p_b	=	2.952131
p_N	=	2.952131
ψ_b	=	0.000000
C_6	=	26.506561
C_1	=	6.274342
C_3	=	8.721514
C_4	=	9.226474
C_5	=	11.577563
C_2	=	8.625431
Z	=	5.303220
m_p	=	1.796404

G3, G4, G5

n_r	=	0.796404
p_x	=	–
m_F	=	–
n_a	=	–
$L_{\min}$	=	12.500000
m_N	=	1.000000
ψ_r	=	0.000000
ϕ_{nr}	=	0.349066

I subroutine

d	=	51.000000
R_{m1}	=	25.612750
ρ_1	=	8.625431
ρ_2	=	17.881130
ρ_{m1}	=	9.045870
ρ_{m2}	=	17.460691
C_ψ	=	1.000000
I	=	0.107

G6 pinion

n_1	=	51.000000
R_{o1}	=	26.612500
R_1	=	25.500000
R_{b1}	=	23.962162
C_4	=	9.226474
x	=	0.112700
Δs_n	=	0.021500
n_c	=	10000.000000
h_{ao}	=	1.456500
x_o	=	0.000000
ρ_{ao}	=	0.436500
δ_{ao}	=	0.009900

J factor pinion

n	=	51.000000
r_n	=	25.500000
r_{nb}	=	23.962162
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	0.385043
x_g	=	0.083165

s_n	=	1.631335
ϕ_{nL}	=	0.338152
r_{nL}	=	25.400616
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r_{no}^s	=	5001.020000
ϕ_{ns}	=	0.349626
$\text{inv}\phi_{ns}$	=	0.014979
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000008
$\text{inv}\phi_n''$	=	0.014910
ϕ''_{ni}	=	0.349111
r_n''	=	25.500422
r_{no}''	=	5000.082737
α_n	=	0.605442
μ_{no}	=	0.000271
κ_s	=	–1.646525
κ_F	=	–2.083025
θ_n	=	0.116255
β_n	=	0.489187
ξ_{nF}	=	1.119150
η_{nF}	=	24.349464
h_F	=	1.051153
y'	=	3.320554
y	=	0.000000
α_{n1}	=	0.605442
ρ_F	=	0.469891
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.238300
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.955832
K_ψ	=	1.000000

G6 gear

n_1	=	104.000000
m_G	=	0.490385
T_{o1}	=	26.612500
R_{o1}	=	52.887000
R_{o2}	=	26.612500
T_1	=	25.500000
R_1	=	52.000000
R_2	=	25.500000
R_{b1}	=	48.864016
C_4	=	17.881130
x	=	–0.112700
Δs_n	=	0.021500
n_c	=	10000.000000
h_{ao}	=	1.456500
x_o	=	0.000000
ρ_{ao}	=	0.436500
δ_{ao}	=	0.009900

J factor gear

n	=	104.000000
r_n	=	52.000000
r_{nb}	=	48.864016
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	0.365937
x_g	=	–0.142235
s_n	=	1.467257
ϕ_{nL}	=	0.336924

Y	=	0.904103
J	=	0.46

r_{nL}	=	51.775013
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r_{no}^s	=	5001.020000
ϕ_{ns}	=	0.349626
$\text{inv}\phi_{ns}$	=	0.014979
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000008
$\text{inv}\phi_n''$	=	0.014894
ϕ''_{ni}	=	0.348988
r_n''	=	51.998536
r_{no}''	=	4999.859214
α_n	=	0.539938
μ_{no}	=	0.000387
κ_s	=	–2.257248
κ_F	=	–2.693748
θ_n	=	0.068205
β_n	=	0.471733
ξ_{nF}	=	1.144259
η_{nF}	=	50.653516
h_F	=	1.121497
y'	=	3.367754
y	=	0.000000
α_{n1}	=	0.539938
ρ_F	=	0.462105
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.288518
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.932315
K_ψ	=	1.000000
Y	=	0.879759
J	=	0.46

Table 3A – Inaccurate spur gears , example 3.1.2

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014910	0.014910	0.014910	–	–	–
ϕ''_{ni}	0.358675	0.349386	0.349112	–	–	–
$\phi''_{n(i+1)}$	0.349386	0.349112	0.349111	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.493311	0.415646	0.413704	0.413704	–
μ_{no}	0.000187	0.000348	0.000424	0.000427	0.000427	–
κ_s	–1.325365	–1.978607	–2.320097	–2.330343	–2.330347	–
κ_F	–1.761865	–2.415107	–2.756597	–2.766843	–2.766847	–
θ_n	0.099917	0.131500	0.146393	0.146832	0.146833	–
β_n	0.685481	0.361811	0.269253	0.266872	0.266871	–
ξ_{nF}	1.179814	1.084906	1.062498	1.061951	1.061951	–
η_{nF}	24.257896	24.425387	24.494373	24.496365	24.496365	–
h_F	2.180642	2.013150	1.944165	1.942173	1.942172	–
y'	8.171188	5.651664	5.408071	5.404076	5.404074	–
y	2.386698	0.438935	0.010502	0.000004	0.000000	–
$\|y\|$	2.386698	0.438935	0.010502	0.000004	0.000000	–
α_{n1}	0.493311	0.415646	0.413704	0.413704	0.413704	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014894	0.014894	0.014894	–	–	–
ϕ''_{ni}	0.358546	0.349263	0.348989	–	–	–
$\phi''_{n(i+1)}$	0.349263	0.348989	0.348988	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.471085	0.368409	0.364454	0.364451	–
μ_{no}	0.000232	0.000456	0.000601	0.000608	0.000608	–
κ_s	–1.641409	–2.556479	–3.220720	–3.254026	–3.254052	–
κ_F	–2.077909	–2.992979	–3.657220	–3.690526	–3.690552	–
θ_n	0.053295	0.074778	0.088748	0.089434	0.089435	–
β_n	0.732103	0.396307	0.279661	0.275020	0.275016	–
ξ_{nF}	1.224450	1.123731	1.093594	1.092425	1.092424	–
η_{nF}	50.535758	50.697888	50.784392	50.788499	50.788502	–
h_F	2.188528	2.026398	1.939895	1.935788	1.935784	–
y'	8.619650	5.572368	5.216194	5.208120	5.208114	–
y	2.709265	0.572150	0.020632	0.000016	0.000000	–
$\|y\|$	2.709265	0.572150	0.020632	0.000016	0.000000	–
α_{n1}	0.471085	0.368409	0.364454	0.364451	0.364451	–

Table 3B – Inaccurate spur gears , example 3.1.2

Input data

Gearset

m_n	=	0.200000
ϕ_n	=	20.0000
ψ	=	0.0000
C_r	=	15.5000
F	=	2.5000
F_1	=	1
F_2	=	2

Pinion

n_1	=	51
n_{c1}	=	10000
R_{o1}	=	5.3225
x_1	=	0.1127
x_{o1}	=	0.0000

Cutter figure 2

h_{ao1}	=	1.4565
ρ_{ao1}	=	0.4365
δ_{ao1}	=	0.0099
Δs_{n1}	=	0.0215

Gear

n_2	=	104
n_{c2}	=	10000
R_{o2}	=	10.5774
x_2	=	–0.1127
x_{o2}	=	0.0000

Cutter figure 2

h_{ao2}	=	1.4565
ρ_{ao2}	=	0.4365
δ_{ao2}	=	0.0099
Δs_{n2}	=	0.0215

G2

ϕ_n	=	0.349066
ψ	=	0.000000
C_r	=	77.500000
F	=	12.500000
R_{o1}	=	26.612500
R_{o2}	=	52.887000
m_G	=	2.039216
R_1	=	25.500000
R_2	=	52.000000
ϕ	=	0.349066
R_{b1}	=	23.962162
R_{b2}	=	48.864016
ϕ_r	=	0.349066
p_b	=	2.952131
p_N	=	2.952131
ψ_b	=	0.000000
C_6	=	26.506561
C_1	=	6.274342
C_3	=	8.721514
C_4	=	9.226474
C_5	=	11.577563
C_2	=	8.625431
Z	=	5.303220
m_p	=	1.796404

G3, G4, G5

n_r	=	0.796404
p_x	=	–
m_F	=	–
n_a	=	–
$L_{\min}$	=	12.500000
m_N	=	1.000000
ψ_r	=	0.000000
ϕ_{nr}	=	0.349066

I subroutine

d	=	51.000000
R_{m1}	=	25.612750
ρ_1	=	8.625431
ρ_2	=	17.881130
ρ_{m1}	=	9.045870
ρ_{m2}	=	17.460691
C_ψ	=	1.000000
I	=	0.107

G6 pinion

n_1	=	51.000000
R_{o1}	=	26.612500
R_1	=	25.500000
R_{b1}	=	23.962162
C_4	=	9.226474
x	=	0.112700
Δs_n	=	0.021500
n_c	=	10000.000000
h_{ao}	=	1.456500
x_o	=	0.000000
ρ_{ao}	=	0.436500
δ_{ao}	=	0.009900

J factor pinion

n	=	51.000000
r_n	=	25.500000
r_{nb}	=	23.962162
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	26.612500
$\tan\phi_{nW}$	=	0.483160
x_g	=	0.083165

s_n	=	1.631335
ϕ_{nL}	=	0.436269
r_{nL}	=	26.438538
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r^s_{no}	=	5001.020000
ϕ_{ns}	=	0.349626
$\text{inv}\phi_{ns}$	=	0.014979
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000008
$\text{inv}\phi''_n$	=	0.014910
ϕ''_{ni}	=	0.349111
r''_n	=	25.500422
r''_{no}	=	5000.082737
α_n	=	0.413704
μ_{no}	=	0.000427
κ_s	=	–2.330347
κ_F	=	–2.766847
θ_n	=	0.146833
β_n	=	0.266871
ξ_{nF}	=	1.061951
η_{nF}	=	24.496365
h_F	=	1.942172
y'	=	5.404074
y	=	0.000000
α_{n1}	=	0.413704
ρ_F	=	0.469891
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.123901
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.485424
K_ψ	=	1.000000

G6 gear

n_1	=	104.000000
m_G	=	0.490385
T_{o1}	=	26.612500
R_{o1}	=	52.887000
R_{o2}	=	26.612500
T_1	=	25.500000
R_1	=	52.000000
R_2	=	25.500000
R_{b1}	=	48.864016
C_4	=	17.881130
x	=	–0.112700
Δs_n	=	0.021500
n_c	=	10000.000000
h_{ao}	=	1.456500
x_o	=	0.000000
ρ_{ao}	=	0.436500
δ_{ao}	=	0.009900

J factor gear

n	=	104.000000
r_n	=	52.000000
r_{nb}	=	48.864016
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	52.887000
$\tan\phi_{nW}$	=	0.414051
x_g	=	–0.142235
s_n	=	1.467257
ϕ_{nL}	=	0.385039

r_{nL}	=	52.724286
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r^s_{no}	=	5001.020000
ϕ_{ns}	=	0.349626
$\text{inv}\phi_{ns}$	=	0.014979
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000008
$\text{inv}\phi''_n$	=	0.014894
ϕ''_{ni}	=	0.348988
r''_n	=	51.998536
r''_{no}	=	4999.859214
α_n	=	0.364451
μ_{no}	=	0.000608
κ_s	=	–3.254052
κ_F	=	–3.690552
θ_n	=	0.089435
β_n	=	0.275016
ξ_{nF}	=	1.092424
η_{nF}	=	50.788502
h_F	=	1.935784
y'	=	5.208114
y	=	0.000000
α_{n1}	=	0.364451
ρ_F	=	0.462105
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.184848
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.513080
K_ψ	=	1.000000
Y	=	0.451108
J	=	0.30

Table 4A – Conventional helical gears, example 3.1.3

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014937	0.014937	0.014937	–	–	–
ϕ''_{ni}	0.358888	0.349589	0.349314	–	–	–
$\phi''_{n(i+1)}$	0.349589	0.349314	0.349313	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.530778	0.484332	0.483773	–	–
μ_{no}	0.000102	0.000174	0.000194	0.000194	–	–
κ_s	–0.801380	–1.119323	–1.216848	–1.218144	–	–
κ_F	–1.210580	–1.528523	–1.626048	–1.627344	–	–
θ_n	0.186208	0.220419	0.229993	0.230119	–	–
β_n	0.599190	0.310360	0.254339	0.253654	–	–
ξ_{nF}	1.157476	1.092056	1.082560	1.082451	–	–
η_{nF}	10.767751	10.903191	10.935968	10.936387	–	–
h_F	2.256396	2.120957	2.088180	2.087760	–	–
y'	7.558439	5.779360	5.652986	5.651830	–	–
y	1.924528	0.268428	0.003165	0.000000	–	–
$\|y\|$	1.924528	0.268428	0.003165	0.000000	–	–
α_{n1}	0.530778	0.484332	0.483773	0.483773	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014902	0.014902	0.014902	–	–	–
ϕ''_{ni}	0.358611	0.349324	0.349050	–	–	–
$\phi''_{n(i+1)}$	0.349324	0.349050	0.349050	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.470846	0.370221	0.366647	0.366645	–
μ_{no}	0.000198	0.000389	0.000510	0.000516	0.000516	–
κ_s	–1.554759	–2.422778	–3.036768	–3.064963	–3.064981	–
κ_F	–1.963959	–2.831978	–3.445968	–3.474163	–3.474181	–
θ_n	0.056632	0.078839	0.092918	0.093552	0.093552	–
β_n	0.728766	0.392006	0.277303	0.273096	0.273093	–
ξ_{nF}	1.235535	1.140587	1.112683	1.111685	1.111685	–
η_{nF}	46.328373	46.482618	46.563564	46.567096	46.567098	–
h_F	2.223231	2.068985	1.988040	1.984507	1.984505	–
y'	8.691103	5.665191	5.332131	5.325511	5.325507	–
y	2.733809	0.570059	0.019054	0.000012	0.000000	–
$\|y\|$	2.733809	0.570059	0.019054	0.000012	0.000000	–
α_{n1}	0.470846	0.370221	0.366647	0.366645	0.366645	–

Table 4B – Conventional helical gears, example 3.1.3

Input data

Gearset

n_n	=	0.166667
ϕ_n	=	20.0000
ψ	=	15.0000
C_r	=	9.3175
F	=	3.7500
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	21
n_{c1}	=	10000
R_{o1}	=	2.0667
x_1	=	0.5343
x_{o1}	=	0.0000

Cutter figure 3

h_{ao1}	=	1.4760
ρ_{ao1}	=	0.4092
δ_{ao1}	=	0.0061
Δs_{n1}	=	0.0240

Gear

n_2	=	86
n_{c2}	=	10000
R_{o2}	=	7.5865
x_2	=	0.0000
x_{o2}	=	0.0000

Cutter figure 3

h_{ao2}	=	1.4760
ρ_{ao2}	=	0.4092
δ_{ao2}	=	0.0061
Δs_{n2}	=	0.0240

G2

ϕ_n	=	0.349066
ψ	=	0.261799
C_r	=	55.904888
F	=	22.499955
R_{o1}	=	12.400175
R_{o2}	=	45.518909
m_G	=	4.095238
R_1	=	10.870400
R_2	=	44.516876
ϕ	=	0.360356
R_{b1}	=	10.172208
R_{b2}	=	41.657612
ϕ_r	=	0.384177
ν	=	3.043517
ρ_N	=	2.952131
ψ_b	=	0.245674
C_6	=	20.952955
C_1	=	2.605900
C_3	=	4.112262
C_4	=	5.649417
C_5	=	7.091582
C_2	=	4.048065
Z	=	4.485682
m_p	=	1.473848

G3, G4, G5

n_r	=	0.473848
p_x	=	12.138182
m_F	=	1.853651
n_a	=	0.853651
$L_{\min}$	=	33.224468
m_N	=	0.677210
ψ_r	=	0.264134
ϕ_{nr}	=	0.372068

_I_ subroutine

d	=	21.943975
R_{m1}	=	11.393077
ρ_1	=	5.131121
ρ_2	=	15.821834
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.242

G6 pinion

n_1	=	21.000000
R_{o1}	=	12.400175
R_1	=	10.870400
R_{b1}	=	10.172208
C_4	=	5.649417
x	=	0.534300
Δs_n	=	0.024000
n_c	=	10000.000000
h_{ao}	=	1.476000
x_o	=	0.000000
ρ_{ao}	=	0.409200
δ_{ao}	=	0.006100

_J_ factor pinion

n	=	23.301719
r_n	=	11.650859
r_{nb}	=	10.948227
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	13.180635
$\tan\phi_{nW}$	=	0.670365
x_g	=	0.501330

s_n	=	1.935735
ϕ_{nL}	=	0.572388
r_{nL}	=	13.024147
n_o	=	11096.056659
r_{no}	=	5548.028330
r_{nbo}	=	5213.441281
r_{no}^s	=	5549.095130
ϕ_{ns}	=	0.349594
$\text{inv}\phi_{ns}$	=	0.014974
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015046
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi_n''$	=	0.014937
ϕ''_{ni}	=	0.349313
r_n''	=	11.651910
r_{no}''	=	5548.528439
α_n	=	0.483773
μ_{no}	=	0.000194
κ_s	=	–1.218144
κ_F	=	–1.627344
θ_n	=	0.230119
β_n	=	0.253654
ξ_{nF}	=	1.082451
η_{nF}	=	10.936387
h_F	=	2.087760
y'	=	5.651830
y	=	0.000000
α_{n1}	=	0.483773
ρ_F	=	0.435535
ω	=	5.236189
C_h	=	1.286597
s_F	=	2.164902
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.472865
K_ψ	=	0.932426

Y	=	0.580609
J	=	0.58

G6 gear

n_1	=	86.000000
m_G	=	0.244186
T_{o1}	=	12.400175
R_{o1}	=	45.518909
R_{o2}	=	12.400175
T_1	=	10.870400
R_1	=	44.516876
R_2	=	10.870400
R_{b1}	=	41.657612
C_4	=	16.904890
x	=	0.000000
Δs_n	=	0.024000
n_c	=	10000.000000
h_{ao}	=	1.476000
x_o	=	0.000000
ρ_{ao}	=	0.409200
δ_{ao}	=	0.006100

_J_ factor gear

n	=	95.426087
r_n	=	47.713044
r_{nb}	=	44.835595
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	48.715077
$\tan\phi_{nW}$	=	0.424901
x_g	=	–0.032970
s_n	=	1.546796
ϕ_{nL}	=	0.393767

r_{nL}	=	48.551604
n_o	=	11096.056659
r_{no}	=	5548.028330
r_{nbo}	=	5213.441281
r_{no}^s	=	5549.095130
ϕ_{ns}	=	0.349594
$\text{inv}\phi_{ns}$	=	0.014974
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015046
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi''_n$	=	0.014902
ϕ''_{ni}	=	0.349050
r_n''	=	47.712763
r_{no}''	=	5547.995640
α_n	=	0.366645
μ_{no}	=	0.000516
κ_s	=	–3.064981
κ_F	=	–3.474181
θ_n	=	0.093552
β_n	=	0.273093
ξ_{nF}	=	1.111685
η_{nF}	=	46.567098
h_F	=	1.984505
y'	=	5.325507
y	=	0.000000
α_{n1}	=	0.366645
ρ_F	=	0.434174
ω	=	5.236189
C_h	=	1.286597
s_F	=	2.223369
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.524664
K_ψ	=	0.932426
Y	=	0.558144
J	=	0.54

Table 5A – Low axial contact ratio (LACR) helical gears, example 3.1.4

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014937	0.014937	0.014937	–	–	–
ϕ''_{ni}	0.358888	0.349589	0.349314	–	–	–
$\phi''_{n(i+1)}$	0.349589	0.349314	0.349313	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.664320	0.651858	0.651758	–	–
μ_{no}	0.000102	0.000130	0.000134	0.000134	–	–
κ_s	–0.801380	–0.919094	–0.934027	–0.934149	–	–
κ_F	–1.210580	–1.328294	–1.343227	–1.343349	–	–
θ_n	0.186208	0.199679	0.201299	0.201312	–	–
β_n	0.599190	0.464641	0.450559	0.450446	–	–
ξ_{nF}	1.157476	1.123738	1.120528	1.120502	–	–
η_{nF}	10.767751	10.825179	10.831698	10.831751	–	–
h_F	1.222488	1.165060	1.158541	1.158488	–	–
y'	4.231279	3.548075	3.492580	3.492145	–	–
y	0.512316	0.044217	0.000348	0.000000	–	–
$\|y\|$	0.512316	0.044217	0.000348	0.000000	–	–
α_{n1}	0.664320	0.651858	0.651758	0.651758	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014902	0.014902	0.014902	–	–	–
ϕ''_{ni}	0.358611	0.349324	0.349050	–	–	–
$\phi''_{n(i+1)}$	0.349324	0.349050	0.349050	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.541421	0.476957	0.474521	0.474518	–
μ_{no}	0.000198	0.000329	0.000383	0.000386	0.000386	–
κ_s	–1.554759	–2.132880	–2.394107	–2.405443	–2.405456	–
κ_F	–1.963959	–2.542080	–2.803307	–2.814643	–2.814656	–
θ_n	0.056632	0.071897	0.078164	0.078431	0.078432	–
β_n	0.728766	0.469524	0.398793	0.396090	0.396087	–
ξ_{nF}	1.235535	1.160451	1.142289	1.141610	1.141609	–
η_{nF}	46.328373	46.439304	46.478542	46.480158	46.480160	–
h_F	1.516735	1.405803	1.366566	1.364950	1.364948	–
y'	6.035126	4.127370	3.848488	3.839332	3.839321	–
y	1.472435	0.266065	0.009375	0.000011	0.000000	–
$\|y\|$	1.472435	0.266065	0.009375	0.000011	0.000000	–
α_{n1}	0.541421	0.476957	0.474521	0.474518	0.474518	–

Table 5B – Low axial contact ratio (LACR) helical gears, example 3.1.4

Input data

Gearset

m_n	=	0.166667
ϕ_n	=	20.0000
ψ	=	15.0000
C_r	=	9.3175
F	=	1.8750
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	21
n_{c1}	=	10000
R_{o1}	=	2.0667
x_1	=	0.5343
x_{o1}	=	0.0000

Cutter figure 3

h_{ao1}	=	1.4760
ρ_{ao1}	=	0.4092
δ_{ao1}	=	0.0061
Δs_{n1}	=	0.0240

Gear

n_2	=	86
n_{c2}	=	10000
R_{o2}	=	7.5865
x_2	=	0.0000
x_{o2}	=	0.0000

Cutter figure 3

h_{ao2}	=	1.4760
ρ_{ao2}	=	0.4092
δ_{ao2}	=	0.0061
Δs_{n2}	=	0.0240

G2

ϕ_n	=	0.349066
ψ	=	0.261799
C_r	=	55.904888
F	=	11.249978
R_{o1}	=	12.400175
R_{o2}	=	45.518909
m_G	=	4.095238
R_1	=	10.870400
R_2	=	44.516876
ϕ	=	0.360356
R_{b1}	=	10.172208
R_{b2}	=	41.657612
ϕ_r	=	0.384177
p_b	=	3.043517
p_N	=	2.952131
ψ_b	=	0.245674
C_6	=	20.952955
C_1	=	2.605900
C_3	=	4.112262
C_4	=	5.649417
C_5	=	7.091582
C_2	=	4.048065
Z	=	4.485682
m_p	=	1.473848

G3, G4, G5

n_r	=	0.473848
p_x	=	12.138182
m_F	=	0.926826
n_a	=	–
L_{min}	=	–
m_N	=	1.000000
ψ_r	=	0.264134
ϕ_{nr}	=	0.372068

_I_ subroutine

d	=	21.943975
R_{m1}	=	11.393077
ρ_1	=	4.048065
ρ_2	=	16.904890
ρ_{m1}	=	5.131121
ρ_{m2}	=	15.821834
C_ψ	=	1.320561
I	=	0.241

G6 pinion

n_1	=	21.000000
R_{o1}	=	12.400175
R_1	=	10.870400
R_{b1}	=	10.172208
C_4	=	5.649417
x	=	0.534300
Δs_n	=	0.024000
n_c	=	10000.000000
h_{ao}	=	1.476000
x_o	=	0.000000
ρ_{ao}	=	0.409200
δ_{ao}	=	0.006100

_J_ factor pinion

n	=	23.301719
r_n	=	11.650859
r_{nb}	=	10.948227
C_{n4}	=	5.670768
r_{n2}	=	47.713044
r_{nb2}	=	44.835595
r_{na2}	=	48.715077
C_{n6}	=	21.769309
C_{n1}	=	2.718636
r_{na}	=	–
$\tan\phi_{nW}$	=	0.517962
x_g	=	0.501330

s_n	=	1.935735
ϕ_{nL}	=	0.419985
r_{nL}	=	11.990239
n_o	=	11096.056659
r_{no}	=	5548.028330
r_{nbo}	=	5213.441281
r_{no}^s	=	5549.095130
ϕ_{ns}	=	0.349594
$\text{inv}\phi_{ns}$	=	0.014974
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015046
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi_n''$	=	0.014937
$\phi''ni$	=	0.349313
r_n''	=	11.651910
r_{no}''	=	5548.528439
α_n	=	0.651758
μ_{no}	=	0.000134
κ_s	=	–0.934149
κ_F	=	–1.343349
θ_n	=	0.201312
β_n	=	0.450446
ξ_{nF}	=	1.120502
η_{nF}	=	10.831751
h_F	=	1.158488
y'	=	3.492145
y	=	0.000000
α_{n1}	=	0.651758
ρ_F	=	0.435535
ω	=	5.236189
C_h	=	1.000000
s_F	=	2.241005
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.900525
K_ψ	=	1.000000

Y	=	0.861107
J	=	0.60

G6 gear

n_1	=	86.000000
m_G	=	0.244186
T_{o1}	=	12.400175
R_{o1}	=	45.518909
R_{o2}	=	12.400175
T_1	=	10.870400
R_1	=	44.516876
R_2	=	10.870400
R_{b1}	=	41.657612
C_4	=	16.904890
x	=	0.000000
Δs_n	=	0.024000
n_c	=	10000.000000
h_{ao}	=	1.476000
x_o	=	0.000000
ρ_{ao}	=	0.409200
δ_{ao}	=	0.006100

_J_ factor gear

n	=	95.426087
r_n	=	47.713044
r_{nb}	=	44.835595
C_{n4}	=	17.382131
r_{n2}	=	11.650859
r_{nb2}	=	10.948227
r_{na2}	=	13.180635
C_{n6}	=	21.769309
C_{n1}	=	14.430000
r_{na}	=	–
$\tan\phi_{nW}$	=	0.387686
x_g	=	–0.032970
s_n	=	1.546796
ϕ_{nL}	=	0.356572

r_{nL}	=	47.845108
n_o	=	11096.056659
r_{no}	=	5548.028330
r_{nbo}	=	5213.441281
r_{no}^s	=	5549.095130
ϕ_{ns}	=	0.349594
$\text{inv}\phi_{ns}$	=	0.014974
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015046
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi''_n$	=	0.014902
ϕ''_{ni}	=	0.349050
r_n''	=	47.712763
r_{no}''	=	5547.995640
α_n	=	0.474518
μ_{no}	=	0.000386
κ_s	=	–2.405456
κ_F	=	–2.814656
θ_n	=	0.078432
β_n	=	0.396087
ξ_{nF}	=	1.141609
η_{nF}	=	46.480160
h_F	=	1.364948
y'	=	3.839321
y	=	0.000000
α_{n1}	=	0.474518
ρ_F	=	0.434174
ω	=	5.236189
C_h	=	1.000000
s_F	=	2.283219
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.796876
K_ψ	=	1.000000
Y	=	0.706122
J	=	0.52

Table 6A – Conventional helical gears, different tools, example 3.1.5

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014923	0.014923	0.014923	–	–	–
ϕ''_{ni}	0.358773	0.349479	0.349204	–	–	–
$\phi''_{n(i+1)}$	0.349479	0.349204	0.349204	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.489843	0.423246	0.422544	–	–
μ_{no}	0.000161	0.000301	0.000356	0.000357	–	–
κ_s	–1.428714	–2.146816	–2.458926	–2.462757	–	–
κ_F	–1.548714	–2.266816	–2.578926	–2.582757	–	–
θ_n	0.108435	0.148603	0.164394	0.164585	–	–
β_n	0.676963	0.341240	0.258852	0.257959	–	–
ξ_{nF}	1.174504	1.122203	1.108587	1.108439	–	–
η_{nF}	20.907868	21.006137	21.050445	21.051005	–	–
h_F	2.243744	2.145476	2.101167	2.100608	–	–
y'	8.228124	6.031137	5.929895	5.929600	–	–
y	2.431867	0.401655	0.004159	0.000000	–	–
$\|y\|$	2.431867	0.401655	0.004159	0.000000	–	–
α_{n1}	0.489843	0.423246	0.422544	0.422544	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.016894	0.016894	0.016894	–	–	–
ϕ''_{ni}	0.373767	0.363741	0.363431	–	–	–
$\phi''_{n(i+1)}$	0.363741	0.363431	0.363430	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.478334	0.388037	0.384580	0.384576	–
μ_{no}	0.039613	0.072843	0.089196	0.089925	0.089926	–
κ_s	–1.549491	–2.268065	–2.662398	–2.680317	–2.680339	–
κ_F	–1.729491	–2.448065	–2.842398	–2.860317	–2.860339	–
θ_n	0.068177	0.091832	0.103473	0.103992	0.103992	–
β_n	0.717222	0.386503	0.284564	0.280588	0.280583	–
ξ_{nF}	1.237266	1.152564	1.124054	1.122925	1.122924	–
η_{nF}	36.071536	36.215014	36.297514	36.301402	36.301406	–
h_F	2.178133	2.034656	1.952155	1.948268	1.948263	–
y'	8.343913	5.576394	5.196610	5.183951	5.183935	–
y	2.562113	0.503535	0.017965	0.000022	0.000000	–
$\|y\|$	2.562113	0.503535	0.017965	0.000022	0.000000	–
α_{n1}	0.478334	0.388037	0.384580	0.384576	0.384576	–

Table 6B – Conventional helical gears, different tools, example 3.1.5

Input data

Gearset

m_n	=	0.083333
ϕ_n	=	20.0000
ψ	=	22.1090
C_r	=	4.2837
F	=	0.8750
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	35
n_{c1}	=	10000
R_{o1}	=	1.6843
x_1	=	0.3498
x_{o1}	=	0.0000

Cutter figure 4

h_{ao1}	=	1.4460
ρ_{ao1}	=	0.1200
δ_{ao1}	=	0.0187
Δs_{n1}	=	0.0240

Gear

n_2	=	59
n_{c2}	=	42
R_{o2}	=	2.7638
x_2	=	0.3523
x_{o2}	=	0.0278

Cutter figure 5

h_{ao2}	=	1.4111
ρ_{ao2}	=	0.1800
δ_{ao2}	=	0.0127
Δs_{n2}	=	0.0240

G2

ϕ_n	=	0.349066
ψ	=	0.385875
C_r	=	51.404606
F	=	10.500042
R_{o1}	=	20.211681
R_{o2}	=	33.165733
m_G	=	1.685714
R_1	=	18.888911
R_2	=	31.841306
ϕ	=	0.374334
R_{b1}	=	17.580881
R_{b2}	=	29.636343
ϕ_r	=	0.406423
p_b	=	3.156112
p_N	=	2.952131
ψ_b	=	0.361494
C_6	=	20.321595
C_1	=	5.433579
C_3	=	7.566551
C_4	=	8.589692
C_5	=	9.971191
C_2	=	6.815079
Z	=	4.537612
m_p	=	1.437722

G3, G4, G5

n_r	=	0.437722
p_x	=	8.347090
m_F	=	1.257928
n_a	=	0.257928
$L_{\min}$	=	15.131716
m_N	=	0.693910
ψ_r	=	0.390502
ϕ_{nr}	=	0.378768

***I* subroutine**

d	=	38.280025
R_{m1}	=	19.225277
ρ_1	=	7.779710
ρ_2	=	12.541885
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.166

G6 pinion

n_1	=	35.000000
R_{o1}	=	20.211681
R_1	=	18.888911
R_{b1}	=	17.580881
C_4	=	8.589692
x	=	0.349800
Δs_n	=	0.024000
n_c	=	10000.000000
h_{ao}	=	1.446000
x_o	=	0.000000
ρ_{ao}	=	0.120000
δ_{ao}	=	0.018700

***J* factor pinion**

n	=	44.012358
r_n	=	22.006179
r_{nb}	=	20.679044
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	23.328949
$\tan\phi_{nW}$	=	0.522216
x_g	=	0.316830

s_n	=	1.801430
ϕ_{nL}	=	0.466382
r_{nL}	=	23.151612
n_o	=	12574.959321
r_{no}	=	6287.479660
r_{nbo}	=	5908.298240
r_{no}^s	=	6288.805660
ϕ_{ns}	=	0.349645
$\text{inv}\phi_{ns}$	=	0.014981
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015029
$\lambda_{ns}/2$	=	0.000031
$\text{inv}\phi_n''$	=	0.014923
ϕ''_{ni}	=	0.349204
r_n''	=	22.007284
r_{no}''	=	6287.795326
α_n	=	0.422544
μ_{no}	=	0.000357
κ_s	=	–2.462757
κ_F	=	–2.582757
θ_n	=	0.164585
β_n	=	0.257959
ξ_{nF}	=	1.108439
η_{nF}	=	21.051005
h_F	=	2.100608
y'	=	5.929600
y	=	0.000000
α_{n1}	=	0.422544
ρ_F	=	0.164496
ω	=	7.910180
C_h	=	1.369671
s_F	=	2.216878
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.693445
K_ψ	=	0.856723

Y	=	0.541648
J	=	0.46

G6 gear

n_1	=	59.000000
m_G	=	0.593220
T_{o1}	=	20.211681
R_{o1}	=	33.165733
R_{o2}	=	20.211681
T_1	=	18.888911
R_1	=	31.841306
R_2	=	18.888911
R_{b1}	=	29.636343
C_4	=	13.506516
x	=	0.352300
Δs_n	=	0.024000
n_c	=	42.000000
h_{ao}	=	1.411100
x_o	=	0.027800
ρ_{ao}	=	0.180000
δ_{ao}	=	0.012700

***J* factor gear**

n	=	74.192260
r_n	=	37.096130
r_{nb}	=	34.858960
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	38.420556
$\tan\phi_{nW}$	=	0.463446
x_g	=	0.319330
s_n	=	1.803250
ϕ_{nL}	=	0.424237

r_{nL}	=	38.249670
n_o	=	52.814829
r_{no}	=	26.407415
r_{nbo}	=	24.814853
r_{no}^s	=	27.666315
ϕ_{ns}	=	0.458011
$\text{inv}\phi_{ns}$	=	0.034963
s_{no}	=	1.591033
$\text{inv}\phi_{npo}$	=	0.045029
$\lambda_{ns}/2$	=	0.003324
$\text{inv}\phi''_n$	=	0.016894
ϕ''_{ni}	=	0.363430
r_n''	=	37.294960
r_{no}''	=	26.548955
α_n	=	0.384576
μ_{no}	=	0.089926
κ_s	=	–2.680339
κ_F	=	–2.860339
θ_n	=	0.103992
β_n	=	0.280583
ξ_{nF}	=	1.122924
η_{nF}	=	36.301406
h_F	=	1.948263
y'	=	5.183935
y	=	0.000000
α_{n1}	=	0.384576
ρ_F	=	0.255092
ω	=	7.910180
C_h	=	1.369671
s_F	=	2.245847
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.657346
K_ψ	=	0.856723
Y	=	0.585809
J	=	0.51

Table 7A – Spur sun and planet gear, example 3.1.6

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014928	0.014928	0.014928	–	–	–
ϕ''_{ni}	0.358812	0.349517	0.349242	–	–	–
$\phi''_{n(i+1)}$	0.349517	0.349242	0.349241	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.602014	0.573238	0.572797	–	–
μ_{no}	0.000151	0.000219	0.000233	0.000233	–	–
κ_s	–1.064773	–1.329401	–1.388073	–1.389022	–	–
κ_F	–1.489773	–1.754401	–1.813073	–1.814022	–	–
θ_n	0.168216	0.192695	0.197730	0.197810	–	–
β_n	0.617182	0.409320	0.375509	0.374987	–	–
ξ_{nF}	1.129153	1.071762	1.063646	1.063524	–	–
η_{nF}	12.941082	13.043537	13.063152	13.063463	–	–
h_F	1.473364	1.370909	1.351294	1.350983	–	–
y'	5.246010	4.090220	3.970709	3.968983	–	–
y	0.962033	0.117700	0.001753	0.000000	–	–
$\|y\|$	0.962033	0.117700	0.001753	0.000000	–	–
α_{n1}	0.602014	0.573238	0.572797	0.572797	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.014928	0.014928	0.014928	–	–	–
ϕ''_{ni}	0.358812	0.349517	0.349242	–	–	–
$\phi''_{n(i+1)}$	0.349517	0.349242	0.349241	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.602014	0.573238	0.572797	–	–
μ_{no}	0.000151	0.000219	0.000233	0.000233	–	–
κ_s	–1.064773	–1.329401	–1.388073	–1.389022	–	–
κ_F	–1.489773	–1.754401	–1.813073	–1.814022	–	–
θ_n	0.168216	0.192695	0.197730	0.197810	–	–
β_n	0.617182	0.409320	0.375509	0.374987	–	–
ξ_{nF}	1.129153	1.071762	1.063646	1.063524	–	–
η_{nF}	12.941082	13.043537	13.063152	13.063463	–	–
h_F	1.473364	1.370909	1.351294	1.350983	–	–
y'	5.246010	4.090220	3.970709	3.968983	–	–
y	0.962033	0.117700	0.001753	0.000000	–	–
$\|y\|$	0.962033	0.117700	0.001753	0.000000	–	–
α_{n1}	0.602014	0.573238	0.572797	0.572797	–	–

Table 7B – Spur sun and planet gear, example 3.1.6

Input data

Gearset

m_n	=	0.200000
ϕ_n	=	20.0000
ψ	=	0.0000
C_r	=	5.7500
F	=	2.5000
F_1	=	1
F_2	=	1

Pinion

n_1	=	28
n_{c1}	=	10000
R_{o1}	=	3.0750
x_1	=	0.4098
x_{o1}	=	0.0000

Cutter figure 6

h_{ao1}	=	1.4975
ρ_{ao1}	=	0.4250
δ_{ao1}	=	0.0245
Δs_{n1}	=	0.0650

Gear

n_2	=	28
n_{c2}	=	10000
R_{o2}	=	3.0750
x_2	=	0.4098
x_{o2}	=	0.0000

Cutter figure 6

h_{ao2}	=	1.4975
ρ_{ao2}	=	0.4250
δ_{ao2}	=	0.0245
Δs_{n2}	=	0.0650

G2

ϕ_n	=	0.349066
ψ	=	0.000000
C_r	=	28.750000
F	=	12.500000
R_{o1}	=	15.375000
R_{o2}	=	15.375000
m_G	=	1.000000
R_1	=	14.000000
R_2	=	14.000000
ϕ	=	0.349066
R_{b1}	=	13.155697
R_{b2}	=	13.155697
ϕ_r	=	0.414845
p_b	=	2.952131
p_N	=	2.952131
ψ_b	=	0.000000
C_6	=	11.587626
C_1	=	3.630348
C_3	=	5.793813
C_4	=	6.582480
C_5	=	7.957278
C_2	=	5.005146
Z	=	4.326929
m_p	=	1.465697

G3, G4, G5

n_r	=	0.465697
p_x	=	–
m_F	=	–
n_a	=	–
$L_{\min}$	=	12.500000
m_N	=	1.000000
ψ_r	=	0.000000
ϕ_{nr}	=	0.414845

***I* subroutine**

d	=	28.750000
R_{m1}	=	14.375000
ρ_1	=	5.005146
ρ_2	=	6.582480
ρ_{m1}	=	5.793813
ρ_{m2}	=	5.793813
C_ψ	=	1.000000
I	=	0.091

G6 pinion

n_1	=	28.000000
R_{o1}	=	15.375000
R_1	=	14.000000
R_{b1}	=	13.155697
C_4	=	6.582480
x	=	0.409800
Δs_n	=	0.065000
n_c	=	10000.000000
h_{ao}	=	1.497500
x_o	=	0.000000
ρ_{ao}	=	0.425000
δ_{ao}	=	0.024500

***J* factor pinion**

n	=	28.000000
r_n	=	14.000000
r_{nb}	=	13.155697
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	0.500352
x_g	=	0.320507

s_n	=	1.804106
ϕ_{nL}	=	0.421015
r_{nL}	=	14.414446
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r_{no}^s	=	5001.072500
ϕ_{ns}	=	0.349655
$\text{inv}\phi_{ns}$	=	0.014983
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi_n''$	=	0.014928
ϕ''_{ni}	=	0.349241
r_n''	=	14.000895
r_{no}''	=	5000.319535
α_n	=	0.572797
μ_{no}	=	0.000233
κ_s	=	–1.389022
κ_F	=	–1.814022
θ_n	=	0.197810
β_n	=	0.374987
ξ_{nF}	=	1.063524
η_{nF}	=	13.063463
h_F	=	1.350983
y'	=	3.968983
y	=	0.000000
α_{n1}	=	0.572797
ρ_F	=	0.463530
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.127047
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.721566
K_ψ	=	1.000000

Y	=	0.634206
J	=	0.37

G6 gear

n_1	=	28.000000
m_G	=	1.000000
T_{o1}	=	15.375000
R_{o1}	=	15.375000
R_{o2}	=	15.375000
T_1	=	14.000000
R_1	=	14.000000
R_2	=	14.000000
R_{b1}	=	13.155697
C_4	=	6.582480
x	=	0.409800
Δs_n	=	0.065000
n_c	=	10000.000000
h_{ao}	=	1.497500
x_o	=	0.000000
ρ_{ao}	=	0.425000
δ_{ao}	=	0.024500

***J* factor gear**

n	=	28.000000
r_n	=	14.000000
r_{nb}	=	13.155697
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	0.500352
x_g	=	0.320507
s_n	=	1.804106
ϕ_{nL}	=	0.421015

r_{nL}	=	14.414446
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r_{no}^s	=	5001.072500
ϕ_{ns}	=	0.349655
$\text{inv}\phi_{ns}$	=	0.014983
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi_n''$	=	0.014928
ϕ''_{ni}	=	0.349241
r_n''	=	14.000895
r_{no}''	=	5000.319535
α_n	=	0.572797
μ_{no}	=	0.000233
κ_s	=	–1.389022
κ_F	=	–1.814022
θ_n	=	0.197810
β_n	=	0.374987
ξ_{nF}	=	1.063524
η_{nF}	=	13.063463
h_F	=	1.350983
y'	=	3.968983
y	=	0.000000
α_{n1}	=	0.572797
ρ_F	=	0.463530
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.127047
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.721566
K_ψ	=	1.000000
Y	=	0.634206
J	=	0.37

Table 8A – Spur planet and ring gear, example 3.1.7

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ_n''	0.014928	0.014928	0.014928	–	–	–
ϕ_{ni}''	0.358812	0.349517	0.349242	–	–	–
$\phi_{n(i+1)}''$	0.349517	0.349242	0.349241	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.661551	0.647725	0.647595	–	–
μ_{no}	0.000151	0.000193	0.000199	0.000199	–	–
κ_s	–1.064773	–1.225493	–1.247761	–1.247975	–	–
κ_F	–1.489773	–1.650493	–1.672761	–1.672975	–	–
θ_n	0.168216	0.183502	0.185507	0.185526	–	–
β_n	0.617182	0.478049	0.462218	0.462068	–	–
ξ_{nF}	1.129153	1.089338	1.085164	1.085125	–	–
η_{nF}	12.941082	13.006523	13.014738	13.014816	–	–
h_F	1.163157	1.097715	1.089501	1.089422	–	–
y'	4.212840	3.485322	3.421318	3.420729	–	–
y	0.521748	0.048189	0.000446	0.000000	–	–
$\|y\|$	0.521748	0.048189	0.000446	0.000000	–	–
α_{n1}	0.661551	0.647725	0.647595	0.647595	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ_n''	–	–	–	–	–	–
ϕ_{ni}''	–	–	–	–	–	–
$\phi_{n(i+1)}''$	–	–	–	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	–	–	–	–	–	–
μ_{no}	–	–	–	–	–	–
κ_s	–	–	–	–	–	–
κ_F	–	–	–	–	–	–
θ_n	–	–	–	–	–	–
β_n	–	–	–	–	–	–
ξ_{nF}	–	–	–	–	–	–
η_{nF}	–	–	–	–	–	–
h_F	–	–	–	–	–	–
y'	–	–	–	–	–	–
y	–	–	–	–	–	–
$\|y\|$	–	–	–	–	–	–
α_{n1}	–	–	–	–	–	–

Table 8B – Spur planet and ring gear, example 3.1.7

Input data

Gearset

m_n	=	0.200000
ϕ_n	=	20.0000
ψ	=	0.0000
C_r	=	5.7500
F	=	2.5000
F_1	=	2
F_2	=	1

Pinion

n_1	=	28
n_{c1}	=	10000
R_{o1}	=	3.0750
x_1	=	0.4098
x_{o1}	=	0.0000

Cutter figure 6

h_{ao1}	=	1.4975
ρ_{ao1}	=	0.4250
δ_{ao1}	=	0.0245
Δs_{n1}	=	0.0650

Gear

n_2	=	85
n_{c2}	=	30
R_{o2}	=	8.4250
x_2	=	0.6678
x_{o2}	=	–0.2229

Cutter figure 7

h_{ao2}	=	1.2629
ρ_{ao2}	=	0.1500
δ_{ao2}	=	0.0000
Δs_{n2}	=	0.0650

G2

ϕ_n	=	0.349066
ψ	=	0.000000
C_r	=	28.750000
F	=	12.500000
R_{o1}	=	15.375000
R_{o2}	=	42.125000
m_G	=	3.035714
R_1	=	14.000000
R_2	=	42.500000
ϕ	=	0.349066
R_{b1}	=	13.155697
R_{b2}	=	39.936936
ϕ_r	=	0.372222
p_b	=	2.952131
p_N	=	2.952131
ψ_b	=	0.000000
C_6	=	10.455989
C_1	=	2.943890
C_3	=	5.136275
C_4	=	5.896021
C_5	=	7.957278
C_2	=	5.005146
Z	=	5.013388
m_p	=	1.698227

G3, G4, G5

n_r	=	0.698227
p_x	=	–
m_F	=	–
n_a	=	–
$L_{\min}$	=	12.500000
m_N	=	1.000000
ψ_r	=	0.000000
ϕ_{nr}	=	0.372222

***I* subroutine**

d	=	28.245614
R_{m1}	=	14.375000
ρ_1	=	5.005146
ρ_2	=	15.461135
ρ_{m1}	=	5.793813
ρ_{m2}	=	16.249802
C_ψ	=	1.000000
I	=	0.244

G6 pinion

n_1	=	28.000000
R_{o1}	=	15.375000
R_1	=	14.000000
R_{b1}	=	13.155697
C_4	=	5.896021
x	=	0.409800
Δs_n	=	0.065000
n_c	=	10000.000000
h_{ao}	=	1.497500
x_o	=	0.000000
ρ_{ao}	=	0.425000
δ_{ao}	=	0.024500

***J* factor pinion**

n	=	28.000000
r_n	=	14.000000
r_{nb}	=	13.155697
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	0.448172
x_g	=	0.320507

s_n	=	1.804106
ϕ_{nL}	=	0.368836
r_{nL}	=	14.104239
n_o	=	10000.000000
r_{no}	=	5000.000000
r_{nbo}	=	4698.463104
r_{no}^s	=	5001.072500
ϕ_{ns}	=	0.349655
$\text{inv}\phi_{ns}$	=	0.014983
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.015061
$\lambda_{ns}/2$	=	–0.000006
$\text{inv}\phi_n''$	=	0.014928
ϕ''_{ni}	=	0.349241
r_n''	=	14.000895
r_{no}''	=	5000.319535
α_n	=	0.647595
μ_{no}	=	0.000199
κ_s	=	–1.247975
κ_F	=	–1.672975
θ_n	=	0.185526
β_n	=	0.462068
ξ_{nF}	=	1.085125
η_{nF}	=	13.014816
h_F	=	1.089422
y'	=	3.420729
y	=	0.000000
α_{n1}	=	0.647595
ρ_F	=	0.463530
ω	=	0.000000
C_h	=	1.000000
s_F	=	2.170249
H	=	0.180000
L	=	0.150000
M	=	0.450000
K_f	=	1.898920
K_ψ	=	1.000000

Y	=	0.825562
J	=	0.43

G6 gear

n_1	=	–
m_G	=	–
T_{o1}	=	–
R_{o1}	=	–
R_{o2}	=	–
T_1	=	–
R_1	=	–
R_2	=	–
R_{b1}	=	–
C_4	=	–
x	=	–
Δs_n	=	–
n_c	=	–
h_{ao}	=	–
x_o	=	–
ρ_{ao}	=	–
δ_{ao}	=	–

***J* factor gear**

n	=	–
r_n	=	–
r_{nb}	=	–
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	–
x_g	=	–
s_n	=	–
ϕ_{nL}	=	–

r_{nL}	=	–
n_o	=	–
r_{no}	=	–
r_{nbo}	=	–
r_{no}^s	=	–
ϕ_{ns}	=	–
$\text{inv}\phi_{ns}$	=	–
s_{no}	=	–
$\text{inv}\phi_{npo}$	=	–
$\lambda_{ns}/2$	=	–
$\text{inv}\phi_n''$	=	–
ϕ''_{ni}	=	–
r_n''	=	–
r_{no}''	=	–
α_n	=	–
μ_{no}	=	–
κ_s	=	–
κ_F	=	–
θ_n	=	–
β_n	=	–
ξ_{nF}	=	–
η_{nF}	=	–
h_F	=	–
y'	=	–
y	=	–
α_{n1}	=	–
ρ_F	=	–
ω	=	–
C_h	=	–
s_F	=	–
H	=	–
L	=	–
M	=	–
K_f	=	–
K_ψ	=	–
Y	=	–
J	=	–

Table 9A – Helical sun and planet gear, example 3.1.8

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.030017	0.030017	0.030017	–	–	–
ϕ''_{ni}	0.451838	0.437112	0.436527	–	–	–
$\phi''_{n(i+1)}$	0.437112	0.436527	0.436526	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.550858	0.516528	0.516292	–	–
μ_{no}	0.000092	0.000150	0.000163	0.000163	–	–
κ_s	–0.755647	–1.020756	–1.081818	–1.082267	–	–
κ_F	–0.845647	–1.110756	–1.171818	–1.172267	–	–
θ_n	0.183552	0.215756	0.222565	0.222614	–	–
β_n	0.601846	0.335102	0.293964	0.293678	–	–
ξ_{nF}	1.203951	1.180769	1.177409	1.177386	–	–
η_{nF}	9.761434	9.808402	9.818742	9.818818	–	–
h_F	2.003974	1.957006	1.946666	1.946590	–	–
y'	6.604027	5.308212	5.241281	5.240928	–	–
y	1.548908	0.182230	0.001237	0.000000	–	–
$\|y\|$	1.548908	0.182230	0.001237	0.000000	–	–
α_{n1}	0.550858	0.516528	0.516292	0.516292	–	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.026131	0.026131	0.026131	–	–	–
ϕ''_{ni}	0.431629	0.418269	0.417773	–	–	–
$\phi''_{n(i+1)}$	0.418269	0.417773	0.417772	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.532078	0.480780	0.479829	0.479829	–
μ_{no}	0.019982	0.033342	0.037361	0.037442	0.037442	–
κ_s	–0.595631	–0.815390	–0.887999	–0.889483	–0.889484	–
κ_F	–0.775631	–0.995390	–1.067999	–1.069483	–1.069484	–
θ_n	0.132923	0.152962	0.158991	0.159112	0.159112	–
β_n	0.652475	0.379116	0.321789	0.320717	0.320716	–
ξ_{nF}	1.208512	1.173198	1.166728	1.166610	1.166610	–
η_{nF}	13.176464	13.239694	13.257422	13.257776	13.257776	–
h_F	1.837279	1.774049	1.756320	1.755966	1.755966	–
y'	6.313286	4.684785	4.524943	4.522425	4.522424	–
y	1.599280	0.240323	0.004303	0.000001	0.000000	–
$\|y\|$	1.599280	0.240323	0.004303	0.000001	0.000000	–
α_{n1}	0.532078	0.480780	0.479829	0.479829	0.479829	–

Table 9B – Helical sun and planet gear, example 3.1.8

Input data

Gearset

m_n	=	0.111111
ϕ_n	=	25.0000
ψ	=	17.7276
C_r	=	2.5500
F	=	1.5000
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	18
n_{c1}	=	10000
R_{o1}	=	1.2154
x_1	=	0.5420
x_{o1}	=	0.0000

Cutter figure 8

h_{ao1}	=	1.1460
ρ_{ao1}	=	0.0900
δ_{ao1}	=	0.0000
Δs_{n1}	=	0.0180

Gear

n_2	=	24
n_{c2}	=	36
R_{o2}	=	1.5457
x_2	=	0.4316
x_{o2}	=	–0.6985

Cutter figure 9

h_{ao2}	=	1.1287
ρ_{ao2}	=	0.1800
δ_{ao2}	=	0.0000
Δs_{n2}	=	0.0180

G2

ϕ_n	=	0.436332
ψ	=	0.309405
C_r	=	22.950023
F	=	13.500014
R_{o1}	=	10.938611
R_{o2}	=	13.911314
m_G	=	1.333333
R_1	=	9.448671
R_2	=	12.598228
ϕ	=	0.455256
R_{b1}	=	8.486309
R_{b2}	=	11.315079
ϕ_r	=	0.530005
p_b	=	2.962281
p_N	=	2.847250
ψ_b	=	0.279592
C_6	=	11.602094
C_1	=	3.509279
C_3	=	4.972326
C_4	=	6.471560
C_5	=	6.901867
C_2	=	3.939586
Z	=	3.392587
m_p	=	1.145262

G3, G4, G5

n_r	=	0.145262
p_x	=	10.317491
m_F	=	1.308459
n_a	=	0.308459
$L_{\min}$	=	15.604712
m_N	=	0.865124
ψ_r	=	0.321240
ϕ_{nr}	=	0.507400

I subroutine

d	=	19.671448
R_{m1}	=	9.988660
ρ_1	=	5.268385
ρ_2	=	6.333708
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.146

G6 pinion

n_1	=	18.000000
R_{o1}	=	10.938611
R_1	=	9.448671
R_{b1}	=	8.486309
C_4	=	6.471560
x	=	0.542000
Δs_n	=	0.018000
n_c	=	10000.000000
h_{ao}	=	1.146000
x_o	=	0.000000
ρ_{ao}	=	0.090000
δ_{ao}	=	0.000000

J factor pinion

n	=	20.828460
r_n	=	10.414230
r_{nb}	=	9.438498
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	11.904170
$\tan\phi_{nW}$	=	0.768580
x_g	=	0.522699

s_n	=	2.058274
ϕ_{nL}	=	0.639785
r_{nL}	=	11.765408
n_o	=	11571.366413
r_{no}	=	5785.683206
r_{nbo}	=	5243.609743
r_{no}^s	=	5786.739206
ϕ_{ns}	=	0.436723
$\text{inv}\phi_{ns}$	=	0.030060
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.030111
$\lambda_{ns}/2$	=	0.000033
$\text{inv}\phi''_n$	=	0.030017
ϕ''_{ni}	=	0.436526
r''_n	=	10.415169
r''_{no}	=	5786.204859
α_n	=	0.516292
μ_{no}	=	0.000163
κ_s	=	–1.082267
κ_F	=	–1.172267
θ_n	=	0.222614
β_n	=	0.293678
ξ_{nF}	=	1.177386
η_{nF}	=	9.818818
h_F	=	1.946590
y'	=	5.240928
y	=	0.000000
α_{n1}	=	0.516292
ρ_F	=	0.116121
ω	=	7.694021
C_h	=	1.363320
s_F	=	2.354772
H	=	0.140000
L	=	0.110000
M	=	0.500000
K_f	=	1.671480
K_ψ	=	0.903789

Y	=	0.801215
J	=	0.55

G6 gear

n_1	=	24.000000
m_G	=	0.750000
T_{o1}	=	10.938611
R_{o1}	=	13.911314
R_{o2}	=	10.938611
T_1	=	9.448671
R_1	=	12.598228
R_2	=	9.448671
R_{b1}	=	11.315079
C_4	=	7.662508
x	=	0.431600
Δs_n	=	0.018000
n_c	=	36.000000
h_{ao}	=	1.128700
x_o	=	–0.698500
ρ_{ao}	=	0.180000
δ_{ao}	=	0.000000

J factor gear

n	=	27.771279
r_n	=	13.885640
r_{nb}	=	12.584663
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	15.198726
$\tan\phi_{nW}$	=	0.677188
x_g	=	0.412299
s_n	=	1.955313
ϕ_{nL}	=	0.576804

r_{nL}	=	15.013743
n_o	=	41.656919
r_{no}	=	20.828460
r_{nbo}	=	18.876995
r_{no}^s	=	21.078660
ϕ_{ns}	=	0.461130
$\text{inv}\phi_{ns}$	=	0.035727
s_{no}	=	0.919365
$\text{inv}\phi_{npo}$	=	0.052045
$\lambda_{ns}/2$	=	0.006783
$\text{inv}\phi''_n$	=	0.026131
ϕ''_{ni}	=	0.417772
r''_n	=	13.768851
r''_{no}	=	20.653277
α_n	=	0.479829
μ_{no}	=	0.037442
κ_s	=	–0.889484
κ_F	=	–1.069484
θ_n	=	0.159112
β_n	=	0.320716
ξ_{nF}	=	1.166610
η_{nF}	=	13.257776
h_F	=	1.755966
y'	=	4.522424
y	=	0.000000
α_{n1}	=	0.479829
ρ_F	=	0.200831
ω	=	7.694021
C_h	=	1.363320
s_F	=	2.333220
H	=	0.140000
L	=	0.110000
M	=	0.500000
K_f	=	1.649675
K_ψ	=	0.903789
Y	=	0.826131
J	=	0.58

Table 10A – Helical planet and ring gear, example 3.1.9

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.026131	0.026131	0.026131	–	–	–
ϕ''_{ni}	0.431629	0.418269	0.417773	–	–	–
$\phi''_{n(i+1)}$	0.418269	0.417773	0.417772	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.532078	0.480780	0.479829	0.479829	–
μ_{no}	0.019982	0.033342	0.037361	0.037442	0.037442	–
κ_s	–0.595631	–0.815390	–0.887999	–0.889483	–0.889484	–
κ_F	–0.775631	–0.995390	–1.067999	–1.069483	–1.069484	–
θ_n	0.132923	0.152962	0.158991	0.159112	0.159112	–
β_n	0.652475	0.379116	0.321789	0.320717	0.320716	–
ξ_{nF}	1.208512	1.173198	1.166728	1.166610	1.166610	–
η_{nF}	13.176464	13.239694	13.257422	13.257776	13.257776	–
h_F	1.837279	1.774049	1.756320	1.755966	1.755966	–
y'	6.313286	4.684785	4.524943	4.522425	4.522424	–
y	1.599280	0.240323	0.004303	0.000001	0.000000	–
$\|y\|$	1.599280	0.240323	0.004303	0.000001	0.000000	–
α_{n1}	0.532078	0.480780	0.479829	0.479829	0.479829	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	–	–	–	–	–	–
ϕ''_{ni}	–	–	–	–	–	–
$\phi''_{n(i+1)}$	–	–	–	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	–	–	–	–	–	–
μ_{no}	–	–	–	–	–	–
κ_s	–	–	–	–	–	–
κ_F	–	–	–	–	–	–
θ_n	–	–	–	–	–	–
β_n	–	–	–	–	–	–
ξ_{nF}	–	–	–	–	–	–
η_{nF}	–	–	–	–	–	–
h_F	–	–	–	–	–	–
y'	–	–	–	–	–	–
y	–	–	–	–	–	–
$\|y\|$	–	–	–	–	–	–
α_{n1}	–	–	–	–	–	–

Table 10B – Helical planet and ring gear, example 3.1.9

Input data

Gearset

m_n	=	0.111111
ϕ_n	=	25.0000
ψ	=	17.7276
C_r	=	2.5500
F	=	1.5000
F_1	=	2
F_2	=	Not required

Pinion

n_1	=	24
n_{c1}	=	36
R_{o1}	=	1.5457
x_1	=	0.4316
x_{o1}	=	–0.6985

Cutter figure 9

h_{ao1}	=	1.1287
ρ_{ao1}	=	0.1800
δ_{ao1}	=	0.0000
Δs_{n1}	=	0.0180

Gear

n_2	=	69
n_{c2}	=	36
R_{o2}	=	3.8846
x_2	=	–0.1972
x_{o2}	=	0.1111

Cutter figure 10

h_{ao2}	=	1.1362
ρ_{ao2}	=	0.1080
δ_{ao2}	=	0.0000
Δs_{n2}	=	0.0180

G2

ϕ_n	=	0.436332
ψ	=	0.309405
C_r	=	22.950023
F	=	13.500014
R_{o1}	=	13.911314
R_{o2}	=	34.961435
m_G	=	2.875000
R_1	=	12.598228
R_2	=	36.219905
ϕ	=	0.455256
R_{b1}	=	11.315079
R_{b2}	=	32.530852
ϕ_r	=	0.391249
p_b	=	2.962281
p_N	=	2.847250
ψ_b	=	0.279592
C_6	=	8.751830
C_1	=	4.056199
C_3	=	4.667643
C_4	=	7.018480
C_5	=	8.092814
C_2	=	5.130533
Z	=	4.036615
m_p	=	1.362671

G3, G4, G5

n_r	=	0.362671
p_x	=	10.317491
m_F	=	1.308459
n_a	=	0.308459
$L_{\min}$	=	17.938456
m_N	=	0.752574
ψ_r	=	0.301137
ϕ_{nr}	=	0.375282

_I_ subroutine

d	=	24.480024
R_{m1}	=	12.961363
ρ_1	=	6.321860
ρ_2	=	15.073690
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.546

G6 pinion

n_1	=	24.000000
R_{o1}	=	13.911314
R_1	=	12.598228
R_{b1}	=	11.315079
C_4	=	7.018480
x	=	0.431600
Δs_n	=	0.018000
n_c	=	36.000000
h_{ao}	=	1.128700
x_o	=	–0.698500
ρ_{ao}	=	0.180000
δ_{ao}	=	0.000000

_J_ factor pinion

n	=	27.771279
r_n	=	13.885640
r_{nb}	=	12.584663
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	15.198726
$\tan\phi_{nW}$	=	0.677188
x_g	=	0.412299

s_n	=	1.955313
ϕ_{nL}	=	0.576804
r_{nL}	=	15.013743
n_o	=	41.656919
r_{no}	=	20.828460
r_{nbo}	=	18.876995
r_{no}^s	=	21.078660
ϕ_{ns}	=	0.461130
$\text{inv}\phi_{ns}$	=	0.035727
s_{no}	=	0.919365
$\text{inv}\phi_{npo}$	=	0.052045
$\lambda_{ns}/2$	=	0.006783
$\text{inv}\phi''_n$	=	0.026131
ϕ''_{ni}	=	0.417772
r''_n	=	13.768851
r''_{no}	=	20.653277
α_n	=	0.479829
μ_{no}	=	0.037442
κ_s	=	–0.889484
κ_F	=	–1.069484
θ_n	=	0.159112
β_n	=	0.320716
ξ_{nF}	=	1.166610
η_{nF}	=	13.257776
h_F	=	1.755966
y'	=	4.522424
y	=	0.000000
α_{n1}	=	0.479829
ρ_F	=	0.200831
ω	=	7.694021
C_h	=	1.363320
s_F	=	2.333220
H	=	0.140000
L	=	0.110000
M	=	0.500000
K_f	=	1.649675
K_ψ	=	0.909652

Y	=	0.885141
J	=	0.71

G6 gear

n_1	=	–
m_G	=	–
T_{o1}	=	–
R_{o1}	=	–
R_{o2}	=	–
T_1	=	–
R_1	=	–
R_2	=	–
R_{b1}	=	–
C_4	=	–
x	=	–
Δs_n	=	–
n_c	=	–
h_{ao}	=	–
x_o	=	–
ρ_{ao}	=	–
δ_{ao}	=	–

_J_ factor gear

n	=	–
r_n	=	–
r_{nb}	=	–
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	–
$\tan\phi_{nW}$	=	–
x_g	=	–
s_n	=	–
ϕ_{nL}	=	–

r_{nL}	=	–
n_o	=	–
r_{no}	=	–
r_{nbo}	=	–
r_{no}^s	=	–
ϕ_{ns}	=	–
$\text{inv}\phi_{ns}$	=	–
s_{no}	=	–
$\text{inv}\phi_{npo}$	=	–
$\lambda_{ns}/2$	=	–
$\text{inv}\phi''_n$	=	–
ϕ''_{ni}	=	–
r''_n	=	–
r''_{no}	=	–
α_n	=	–
μ_{no}	=	–
κ_s	=	–
κ_F	=	–
θ_n	=	–
β_n	=	–
ξ_{nF}	=	–
η_{nF}	=	–
h_F	=	–
y'	=	–
y	=	–
α_{n1}	=	–
ρ_F	=	–
ω	=	–
C_h	=	–
s_F	=	–
H	=	–
L	=	–
M	=	–
K_f	=	–
K_ψ	=	–
Y	=	–
J	=	–

Table 11A – Conventional double helical gears, example 3.1.10

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.008232	0.008232	0.008232	–	–	–
ϕ''_{ni}	0.294824	0.288120	0.287952	–	–	–
$\phi''_{n(i+1)}$	0.288120	0.287952	0.287952	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.471112	0.373218	0.370062	0.370060	–
μ_{no}	0.000112	0.000220	0.000287	0.000289	0.000289	–
κ_s	–1.432158	–2.230837	–2.776564	–2.799123	–2.799134	–
κ_F	–1.842358	–2.641037	–3.186764	–3.209323	–3.209334	–
θ_n	0.067420	0.093796	0.109961	0.110616	0.110616	–
β_n	0.717978	0.377315	0.263257	0.259446	0.259444	–
ξ_{nF}	1.102763	1.006879	0.979597	0.978719	0.978718	–
η_{nF}	35.669413	35.829899	35.912873	35.916155	35.916156	–
h_F	2.090416	1.929930	1.846956	1.843674	1.843673	–
y'	8.111022	5.340462	5.055824	5.050907	5.050904	–
y	2.549184	0.522796	0.015958	0.000008	0.000000	–
$\|y\|$	2.549184	0.522796	0.015958	0.000008	0.000000	–
α_{n1}	0.471112	0.373218	0.370062	0.370060	0.370060	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.008231	0.008231	0.008231	–	–	–
ϕ''_{ni}	0.294812	0.288109	0.287940	–	–	–
$\phi''_{n(i+1)}$	0.288109	0.287940	0.287940	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.439167	0.294720	0.284233	0.284201	–
μ_{no}	0.000116	0.000246	0.000381	0.000396	0.000396	–
κ_s	–1.475162	–2.452773	–3.589001	–3.717446	–3.717857	–
κ_F	–1.885362	–2.862973	–3.999201	–4.127646	–4.128057	–
θ_n	0.009408	0.013803	0.018337	0.018837	0.018839	–
β_n	0.775990	0.425365	0.276383	0.265396	0.265362	–
ξ_{nF}	1.173481	1.087974	1.062335	1.060625	1.060620	–
η_{nF}	266.440538	266.566018	266.636630	266.642786	266.642806	–
h_F	2.153577	2.028096	1.957484	1.951328	1.951308	–
y'	8.818923	5.189245	4.588415	4.560704	4.560622	–
y	3.053383	0.749573	0.048117	0.000147	0.000000	–
$\|y\|$	3.053383	0.749573	0.048117	0.000147	0.000000	–
α_{n1}	0.439167	0.294720	0.284233	0.284201	0.284201	–

Table 11B – Conventional double helical gears, example 3.1.10

Input data

Gearset

m_n	=	0.204788
ϕ_n	=	16.5000
ψ	=	34.7564
C_r	=	42.1250
F	=	20.0000
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	41
n_{c1}	=	10000
R_{o1}	=	5.3070
x_1	=	0.0166
x_{o1}	=	0.0000

Cutter figure 11

h_{ao1}	=	1.3492
ρ_{ao1}	=	0.4102
δ_{ao1}	=	0.0114
Δs_{n1}	=	0.0537

Gear

n_2	=	297
n_{c2}	=	10000
R_{o2}	=	37.2120
x_2	=	–0.0166
x_{o2}	=	0.0000

Cutter figure 11

h_{ao2}	=	1.3492
ρ_{ao2}	=	0.4102
δ_{ao2}	=	0.0147
Δs_{n2}	=	0.0537

G2

ϕ_n	=	0.287979
ψ	=	0.606614
C_r	=	205.700529
F	=	97.661972
R_{o1}	=	25.914604
R_{o2}	=	181.709866
m_G	=	7.243902
R_1	=	24.951823
R_2	=	180.748570
ϕ	=	0.346033
R_{b1}	=	23.472815
R_{b2}	=	170.034781
ϕ_r	=	0.346035
p_b	=	3.597172
p_N	=	3.012221
ψ_b	=	0.578313
C_6	=	69.767600
C_1	=	5.684462
C_3	=	8.462934
C_4	=	9.281634
C_5	=	10.981515
C_2	=	7.384343
Z	=	5.297053
m_p	=	1.472560

G3, G4, G5

n_r	=	0.472560
p_x	=	5.510710
m_F	=	17.722212
n_a	=	0.722212
$L_{\min}$	=	170.776393
m_N	=	0.571870
ψ_r	=	0.606614
ϕ_{nr}	=	0.287981

I subroutine

d	=	49.903679
R_{m1}	=	24.952634
ρ_1	=	8.465276
ρ_2	=	61.302324
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.245

G6 pinion

n_1	=	41.000000
R_{o1}	=	25.914604
R_1	=	24.951823
R_{b1}	=	23.472815
C_4	=	9.281634
x	=	0.016600
Δs_n	=	0.053700
n_c	=	10000.000000
h_{ao}	=	1.349200
x_o	=	0.000000
ρ_{ao}	=	0.410200
δ_{ao}	=	0.011400

J–Factor Pinion

n	=	73.931430
r_n	=	36.965715
r_{nb}	=	35.443457
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	37.928497
$\tan\phi_{nW}$	=	0.380974
x_g	=	–0.074044

s_n	=	1.526931
ϕ_{nL}	=	0.352087
r_{nL}	=	37.759829
n_o	=	18032.056062
r_{no}	=	9016.028031
r_{nbo}	=	8644.745606
r_{no}^s	=	9016.967031
ϕ_{ns}	=	0.288331
$\text{inv}\phi_{ns}$	=	0.008265
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.008321
$\lambda_{ns}/2$	=	0.000010
$\text{inv}\phi_n''$	=	0.008232
ϕ''_{ni}	=	0.287952
r_n''	=	36.965413
r_{no}''	=	9015.954286
α_n	=	0.370060
μ_{no}	=	0.000289
κ_s	=	–2.799134
κ_F	=	–3.209334
θ_n	=	0.110616
β_n	=	0.259444
ξ_{nF}	=	0.978718
η_{nF}	=	35.916156
h_F	=	1.843673
y'	=	5.050904
y	=	0.000000
α_{n1}	=	0.370060
ρ_F	=	0.437314
ω	=	11.148723
C_h	=	1.459288
s_F	=	1.957437
H	=	0.210000
L	=	0.180000
M	=	0.420000
K_f	=	1.553020
K_ψ	=	0.674999

Y	=	0.385039
J	=	0.43

G6 gear

n_1	=	297.000000
m_G	=	0.138047
T_{o1}	=	25.914604
R_{o1}	=	181.709866
R_{o2}	=	25.914604
T_1	=	24.951823
R_1	=	180.748570
R_2	=	24.951823
R_{b1}	=	170.034781
C_4	=	62.383256
x	=	–0.016600
Δs_n	=	0.053700
n_c	=	10000.000000
h_{ao}	=	1.349200
x_o	=	0.000000
ρ_{ao}	=	0.410200
δ_{ao}	=	0.014700

J factor gear

n	=	535.552065
r_n	=	267.776033
r_{nb}	=	256.748945
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	268.737328
$\tan\phi_{nW}$	=	0.309138
x_g	=	–0.107244
s_n	=	1.507262
ϕ_{nL}	=	0.298089

r_{nL}	=	268.594114
n_o	=	18032.056062
r_{no}	=	9016.028031
r_{nbo}	=	8644.745606
r_{no}^s	=	9016.967031
ϕ_{ns}	=	0.288331
$\text{inv}\phi_{ns}$	=	0.008265
s_{no}	=	1.570796
$\text{inv}\phi_{npo}$	=	0.008321
$\lambda_{ns}/2$	=	0.000010
$\text{inv}\phi_n''$	=	0.008231
ϕ''_{ni}	=	0.287940
r_n''	=	267.772939
r_{no}''	=	9015.923873
α_n	=	0.284201
μ_{no}	=	0.000396
κ_s	=	–3.717857
κ_F	=	–4.128057
θ_n	=	0.018839
β_n	=	0.265362
ξ_{nF}	=	1.060620
η_{nF}	=	266.642806
h_F	=	1.951308
y'	=	4.560622
y	=	0.000000
α_{n1}	=	0.284201
ρ_F	=	0.414368
ω	=	11.148723
C_h	=	1.459288
s_F	=	2.121239
H	=	0.210000
L	=	0.180000
M	=	0.420000
K_f	=	1.599592
K_ψ	=	0.674999
Y	=	0.413301
J	=	0.45

Table 12A – Herringbone gears, example 3.1.11

Pinion: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.010523	0.010523	0.010523	–	–	–
ϕ''_{ni}	0.319708	0.312094	0.311891	–	–	–
$\phi''_{n(i+1)}$	0.312094	0.311891	0.311891	–	–	–
Pinion: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.515048	0.450400	0.448196	0.448194	–
μ_{no}	0.056376	0.094599	0.107977	0.108476	0.108477	–
κ_s	–2.087997	–2.844041	–3.136649	–3.147753	–3.147765	–
κ_F	–2.407597	–3.163641	–3.456249	–3.467353	–3.467365	–
θ_n	0.143301	0.186984	0.202274	0.202845	0.202845	–
β_n	0.642097	0.328064	0.248126	0.245352	0.245348	–
ξ_{nF}	1.156529	1.020337	0.988867	0.987829	0.987828	–
η_{nF}	19.936168	20.203563	20.310241	20.314362	20.314366	–
h_F	2.350755	2.083359	1.976681	1.972560	1.972556	–
y'	8.726844	6.154200	5.765165	5.752152	5.752137	–
y	2.359300	0.397862	0.012703	0.000014	0.000000	–
$\|y\|$	2.359300	0.397862	0.012703	0.000014	0.000000	–
α_{n1}	0.515048	0.450400	0.448196	0.448194	0.448194	–
Gear: iteration for generating pressure angle						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
inv ϕ''_n	0.010061	0.010061	0.010061	–	–	–
ϕ''_{ni}	0.315012	0.307579	0.307383	–	–	–
$\phi''_{n(i+1)}$	0.307579	0.307383	0.307383	–	–	–
Gear: iteration for critical point						
Variable	**1**	**2**	**3**	**4**	**5**	**6**
α_n	0.785398	0.454137	0.328092	0.318479	0.318426	–
μ_{no}	0.057662	0.109425	0.144197	0.147455	0.147474	–
κ_s	–2.135593	–3.184326	–3.977492	–4.053652	–4.054079	–
κ_F	–2.455193	–3.503926	–4.297092	–4.373252	–4.373679	–
θ_n	0.024717	0.034817	0.041602	0.042238	0.042241	–
β_n	0.760681	0.419320	0.286490	0.276241	0.276185	–
ξ_{nF}	1.343689	1.197141	1.132036	1.126785	1.126756	–
η_{nF}	124.597297	124.825383	125.004924	125.023094	125.023196	–
h_F	2.413627	2.185541	2.006000	1.987830	1.987728	–
y'	9.812936	5.960566	5.188472	5.131782	5.131468	–
y	3.250645	0.751299	0.049876	0.000272	0.000000	–
$\|y\|$	3.250645	0.751299	0.049876	0.000272	0.000000	–
α_{n1}	0.454137	0.328092	0.318479	0.318426	0.318426	–

Table 12B – Herringbone gears, example 3.1.11

Input data

Gearset

m_n	=	0.108253
ϕ_n	=	17.4952
ψ	=	30.0000
C_r	=	12.0000
F	=	7.0000
F_1	=	1
F_2	=	Not required

Pinion

n_1	=	28
n_{c1}	=	32
R_{o1}	=	1.8500
x_1	=	0.0000
x_{o1}	=	0.1371

Cutter figure 12

h_{ao1}	=	1.7520
ρ_{ao1}	=	0.3196
δ_{ao1}	=	0.0000
Δs_{n1}	=	0.0250

Gear

n_2	=	164
n_{c2}	=	32
R_{o2}	=	10.3500
x_2	=	0.0000
x_{o2}	=	0.1371

Cutter figure 12

h_{ao2}	=	1.7520
ρ_{ao2}	=	0.3196
δ_{ao2}	=	0.0000
Δs_{n2}	=	0.0250

G2

ϕ_n	=	0.305349
ψ	=	0.523599
C_r	=	110.851431
F	=	64.663335
R_{o1}	=	17.089596
R_{o2}	=	95.609360
m_G	=	5.857143
R_1	=	16.165808
R_2	=	94.685444
ϕ	=	0.349065
R_{b1}	=	15.190894
R_{b2}	=	88.975239
ϕ_r	=	0.349070
p_b	=	3.408829
p_N	=	2.996269
ψ_b	=	0.497092
C_6	=	37.913803
C_1	=	2.920140
C_3	=	5.529096
C_4	=	6.328969
C_5	=	7.828857
C_2	=	4.420028
Z	=	4.908717
m_p	=	1.440001

G3, G4, G5

n_r	=	0.440001
p_x	=	6.283185
m_F	=	10.291489
n_a	=	0.291489
$L_{\min}$	=	105.019605
m_N	=	0.615726
ψ_r	=	0.523599
ϕ_{nr}	=	0.305353

_I_ subroutine

d	=	32.331667
R_{m1}	=	16.165834
ρ_1	=	5.529096
ρ_2	=	32.384707
ρ_{m1}	=	–
ρ_{m2}	=	–
C_ψ	=	1.000000
I	=	0.223

G6 Pinion

n_1	=	28.000000
R_{o1}	=	17.089596
R_1	=	16.165808
R_{b1}	=	15.190894
C_4	=	6.328969
x	=	0.000000
Δs_n	=	0.025000
n_c	=	32.000000
h_{ao}	=	1.752000
x_o	=	0.137100
ρ_{ao}	=	0.319600
δ_{ao}	=	0.000000

_J_ factor pinion

n	=	43.108820
r_n	=	21.554410
r_{nb}	=	20.557349
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	22.478198
$\tan\phi_{nW}$	=	0.442276
x_g	=	–0.039657

s_n	=	1.545796
ϕ_{nL}	=	0.396560
r_{nL}	=	22.286922
n_o	=	49.267223
r_{no}	=	24.633611
r_{nbo}	=	23.494113
r_{no}^s	=	26.203111
ϕ_{ns}	=	0.458730
$\text{inv}\phi_{ns}$	=	0.035138
s_{no}	=	1.657226
$\text{inv}\phi_{npo}$	=	0.043495
$\lambda_{ns}/2$	=	–0.005246
$\text{inv}\phi_n''$	=	0.010523
ϕ''_{ni}	=	0.311891
r_n''	=	21.599415
r_{no}''	=	24.685046
α_n	=	0.448194
μ_{no}	=	0.108477
κ_s	=	–3.147765
κ_F	=	–3.467365
θ_n	=	0.202845
β_n	=	0.245348
ξ_{nF}	=	0.987828
η_{nF}	=	20.314366
h_F	=	1.972556
y'	=	5.752137
y	=	0.000000
α_{n1}	=	0.448194
ρ_F	=	0.496358
ω	=	9.846530
C_h	=	1.424385
s_F	=	1.975656
H	=	0.200000
L	=	0.170000
M	=	0.430000
K_f	=	1.465549
K_ψ	=	0.750000

Y	=	0.404568
J	=	0.45

G6 gear

n_1	=	164.000000
m_G	=	0.170732
T_{o1}	=	17.089596
R_{o1}	=	95.609360
R_{o2}	=	17.089596
T_1	=	16.165808
R_1	=	94.685444
R_2	=	16.165808
R_{b1}	=	88.975239
C_4	=	33.493775
x	=	0.000000
Δs_n	=	0.025000
n_c	=	32.000000
h_{ao}	=	1.752000
x_o	=	0.137100
ρ_{ao}	=	0.319600
δ_{ao}	=	0.000000

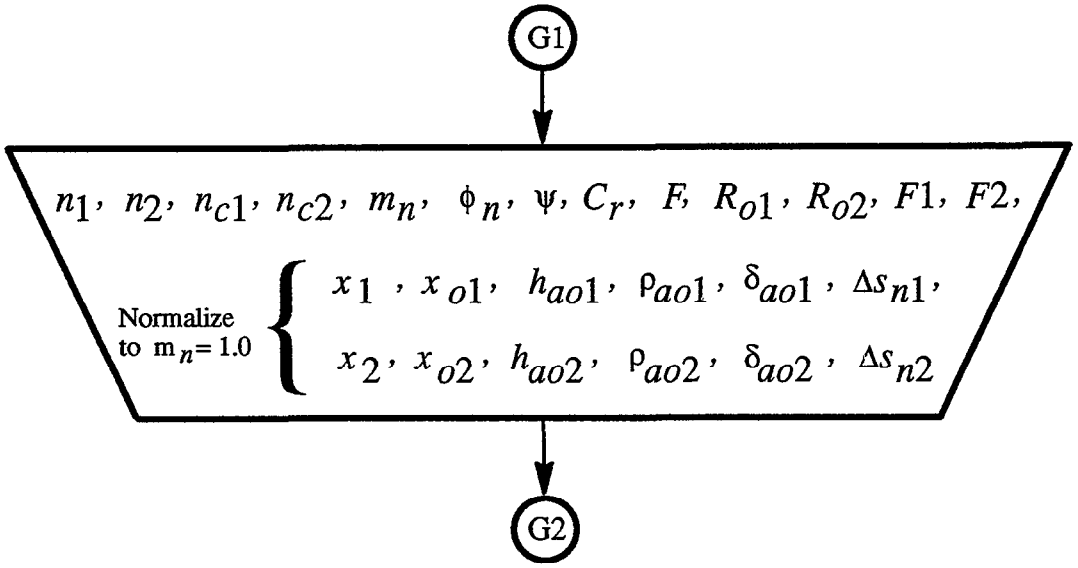
_J_ factor gear

n	=	252.494518
r_n	=	126.247259
r_{nb}	=	120.407331
C_{n4}	=	–
r_{n2}	=	–
r_{nb2}	=	–
r_{na2}	=	–
C_{n6}	=	–
C_{n1}	=	–
r_{na}	=	127.171174
$\tan\phi_{nW}$	=	0.339860
x_g	=	–0.039657
s_n	=	1.545796
ϕ_{nL}	=	0.323880

r_{nL}	=	127.010924
n_o	=	49.267223
r_{no}	=	24.633611
r_{nbo}	=	23.494113
r_{no}^s	=	26.203111
ϕ_{ns}	=	0.458730
$\text{inv}\phi_{ns}$	=	0.035138
s_{no}	=	1.657226
$\text{inv}\phi_{npo}$	=	0.043495
$\lambda_{ns}/2$	=	–0.005246
$\text{inv}\phi_n''$	=	0.010061
ϕ''_{ni}	=	0.307383
r_n''	=	126.328531
r_{no}''	=	24.649469
α_n	=	0.318426
μ_{no}	=	0.147474
κ_s	=	–4.054079
κ_F	=	–4.373679
θ_n	=	0.042241
β_n	=	0.276185
ξ_{nF}	=	1.126756
η_{nF}	=	125.023196
h_F	=	1.987728
y'	=	5.131468
y	=	0.000000
α_{n1}	=	0.318426
ρ_F	=	0.428434
ω	=	9.846530
C_h	=	1.424385
s_F	=	2.253512
H	=	0.200000
L	=	0.170000
M	=	0.430000
K_f	=	1.599594
K_ψ	=	0.750000
Y	=	0.503089
J	=	0.51

4 Flow chart

A flow chart has been developed to assist in the calculation of geometry factors, I and J. This chart organizes an efficient method for programming the procedures presented in AGMA 908–B89. Its format separates input data, basic geometry calculations common to both factors, subroutines for intermediate factors, and subroutines for the final calculation of each geometry factor. Decision flags identify alternate routines for variables based on the specific gear geometry. Each step in the chart identifies a variable and, where applicable, the equation number from AGMA 908–B89 needed to calculate the variable. The numerical value tables in clause 3 are formatted to follow with the flow chart procedures.



For hobs, let $n_{c1} = n_{c2} = 10\,000$, $x_{o1} = x_{o2} = 0.0$

If $s_{no} = \pi/2$, $x_{o1} = x_{o2} = 0.0$

For metric units enter m_n, C_r, F, R_{o1}, and R_{o2} in millimeters.

For inch units enter $m_n = 1/P_{nd}$ and C_r, F, R_{o1}, R_{o2} in inches.

For an internal gear $R_{o2} = D_i/2$.

$F1$= Gear type code:
- 1 = External gear;
- 2 = Internal gear.

$F2$= Spur gear load sharing code:
- 1 = Load at HPSTC;
- 2 = Load at tip (inaccurate).

For protuberance hobs, reduce protuberance:

$$\delta_{ao} = \delta_o - u_s \cos \phi_n$$

Where

δ_o is the protuberance on the cutter;

u_s is the stock allowance per side of tooth for finishing.

Figure 1 – Flow chart for I and J subroutines for AGMA 908–B89

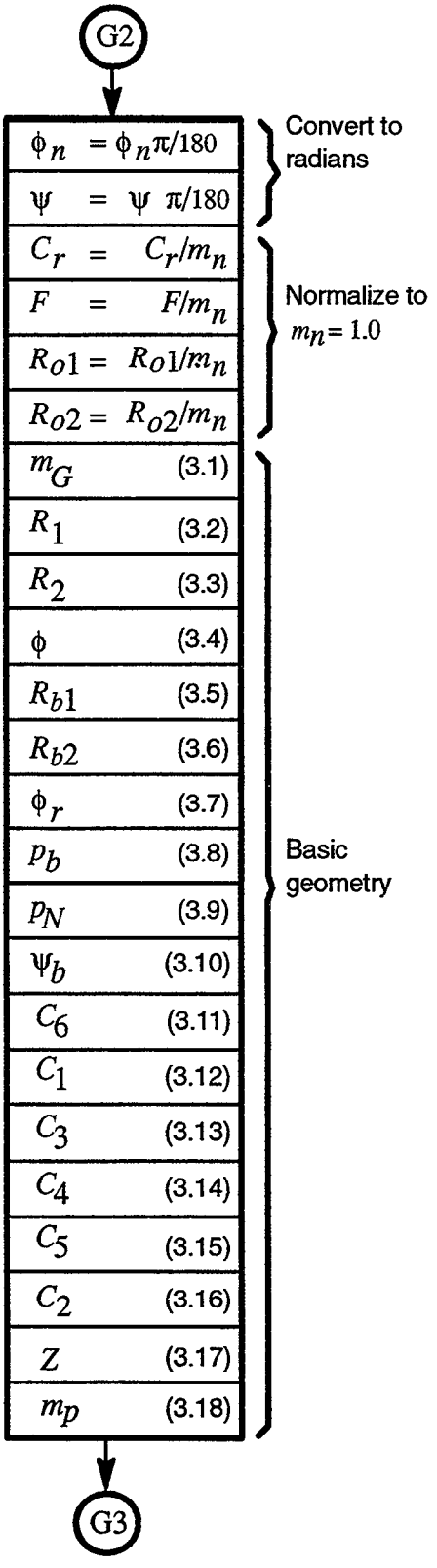


Figure 1 *(continued)*

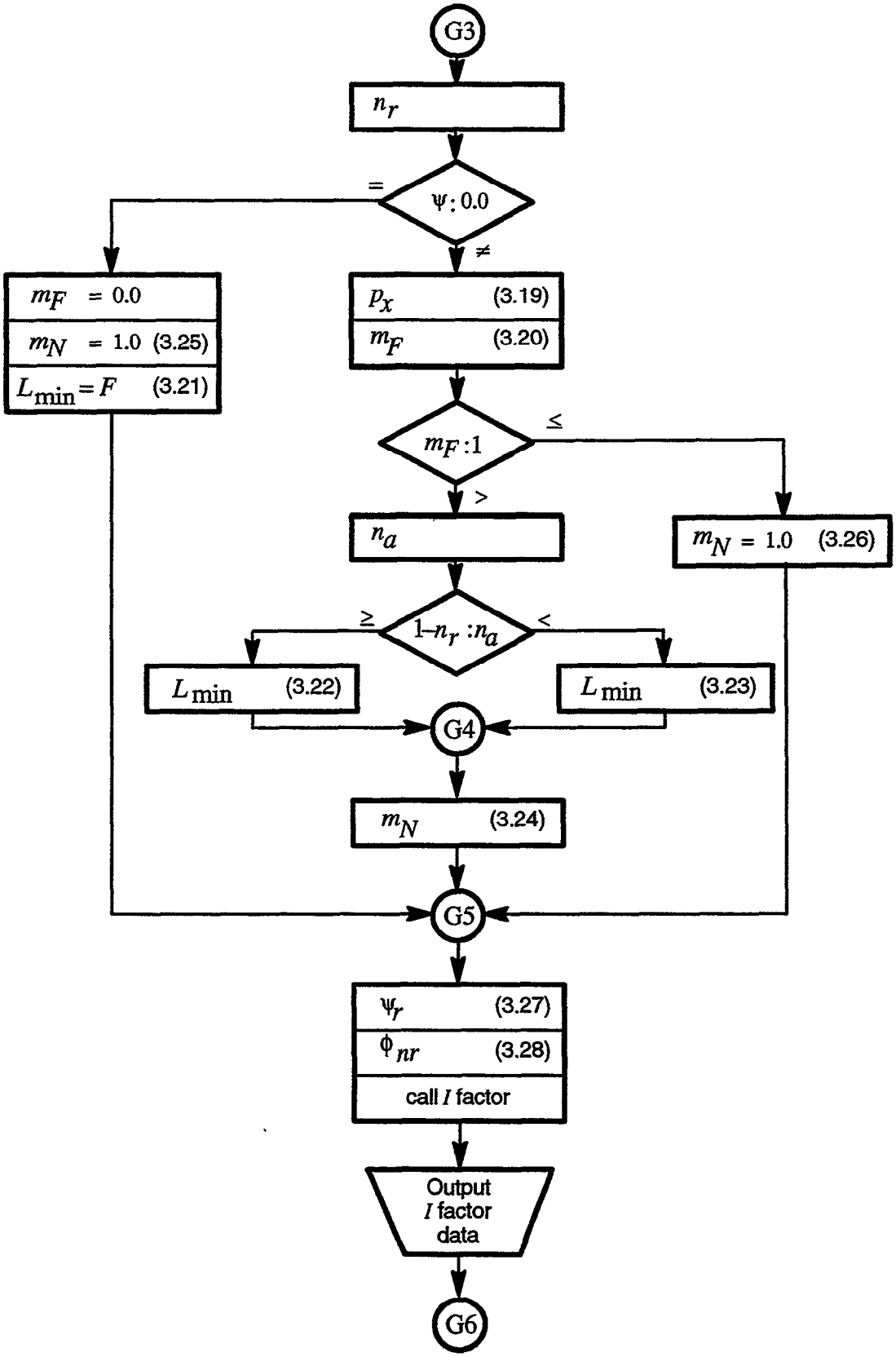


Figure 1 *(continued)*

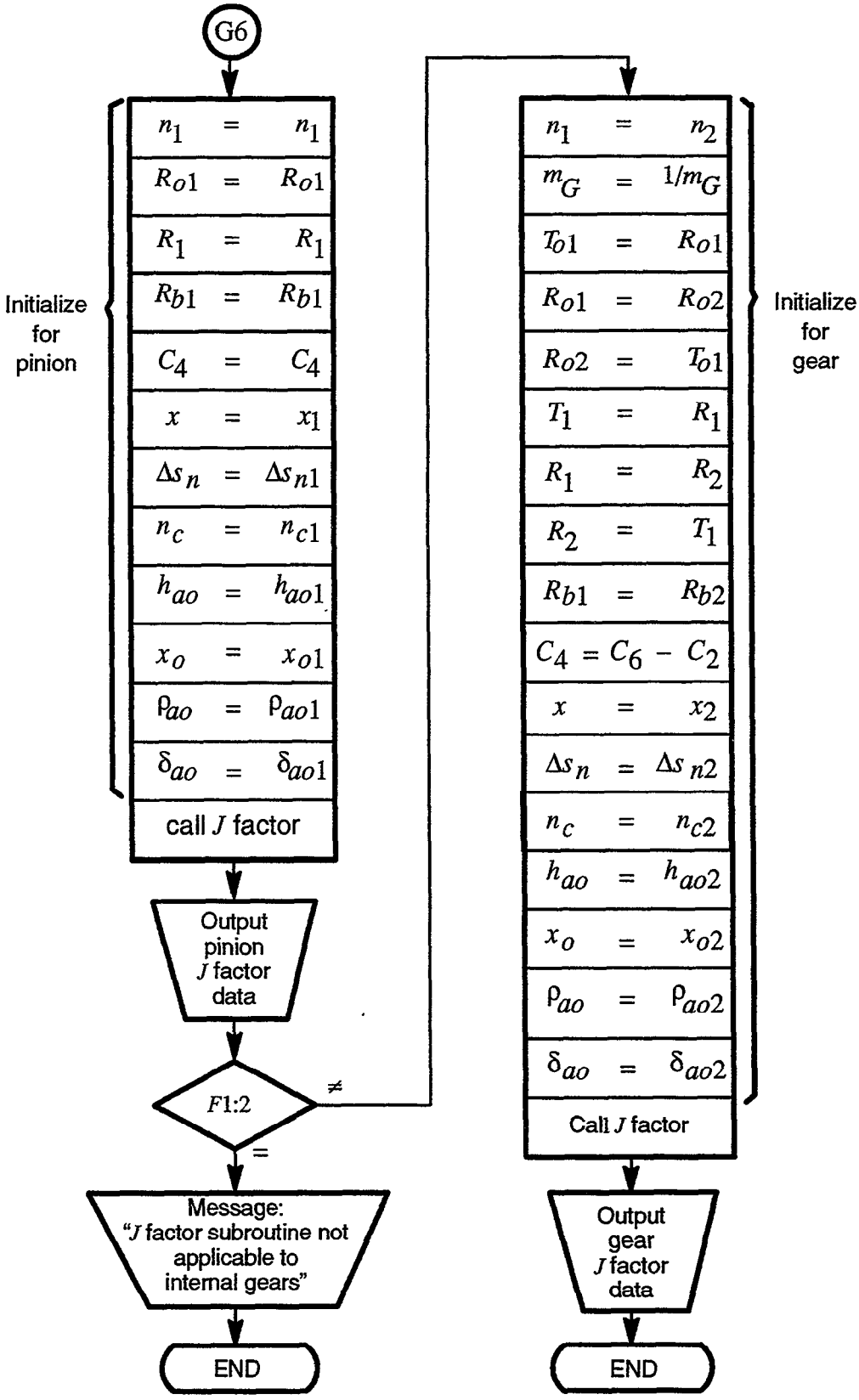


Figure 1 *(continued)*

I Factor subroutine

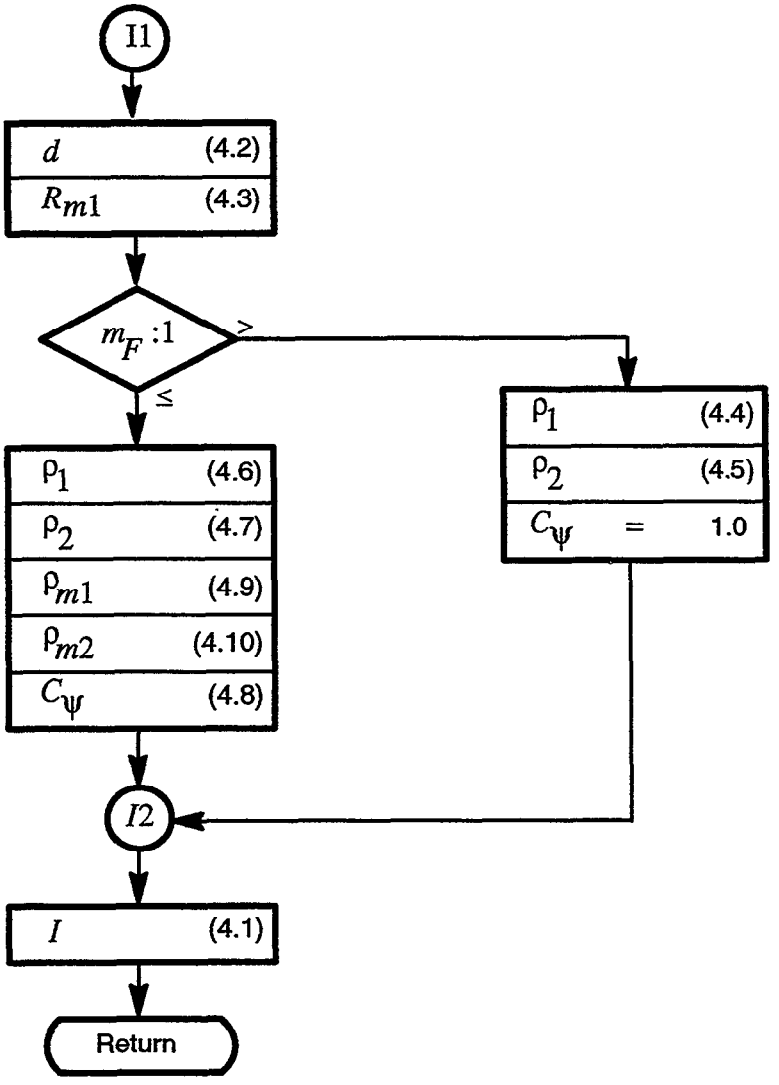


Figure 1 *(continued)*

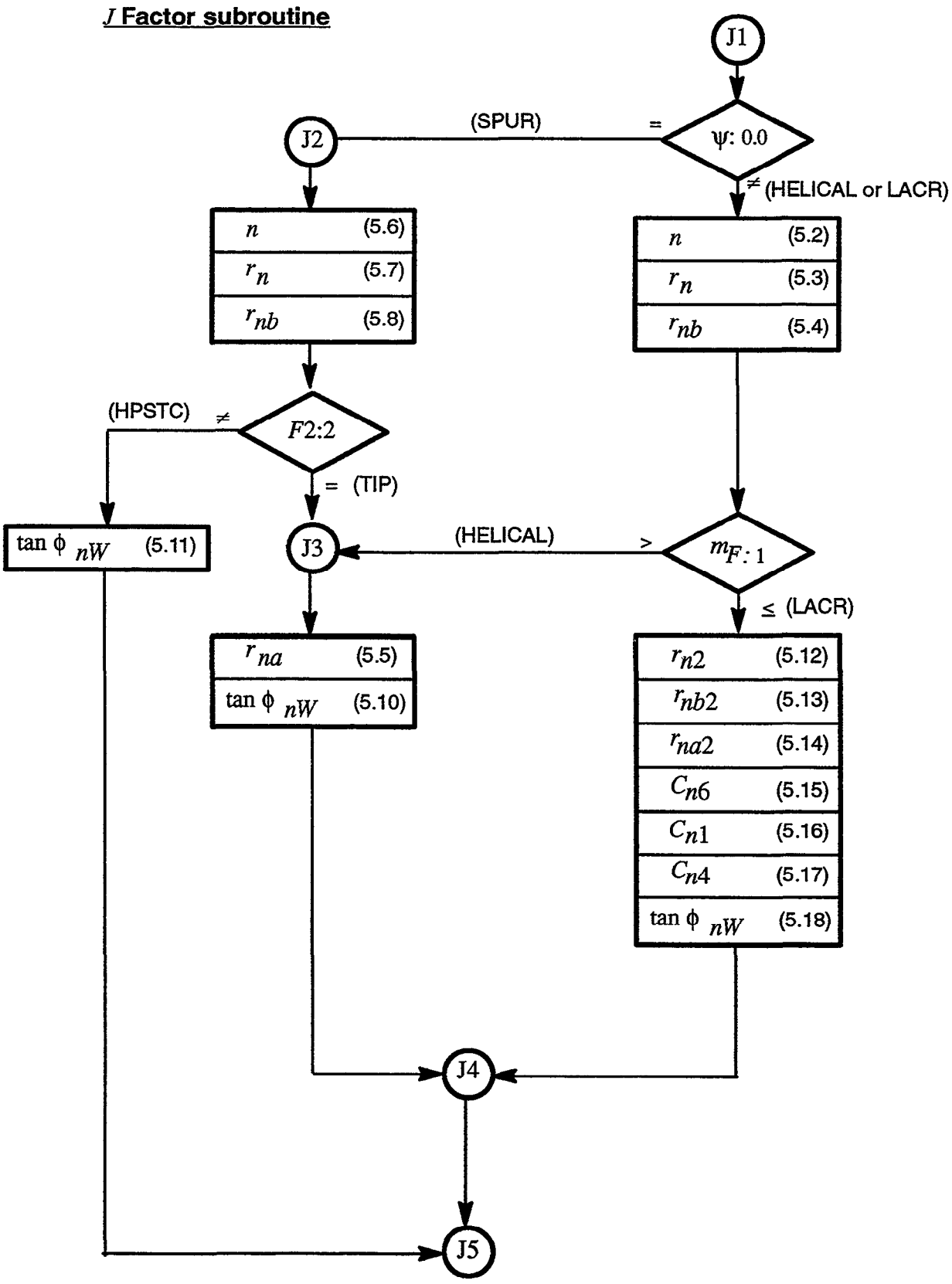


Figure 1 *(continued)*

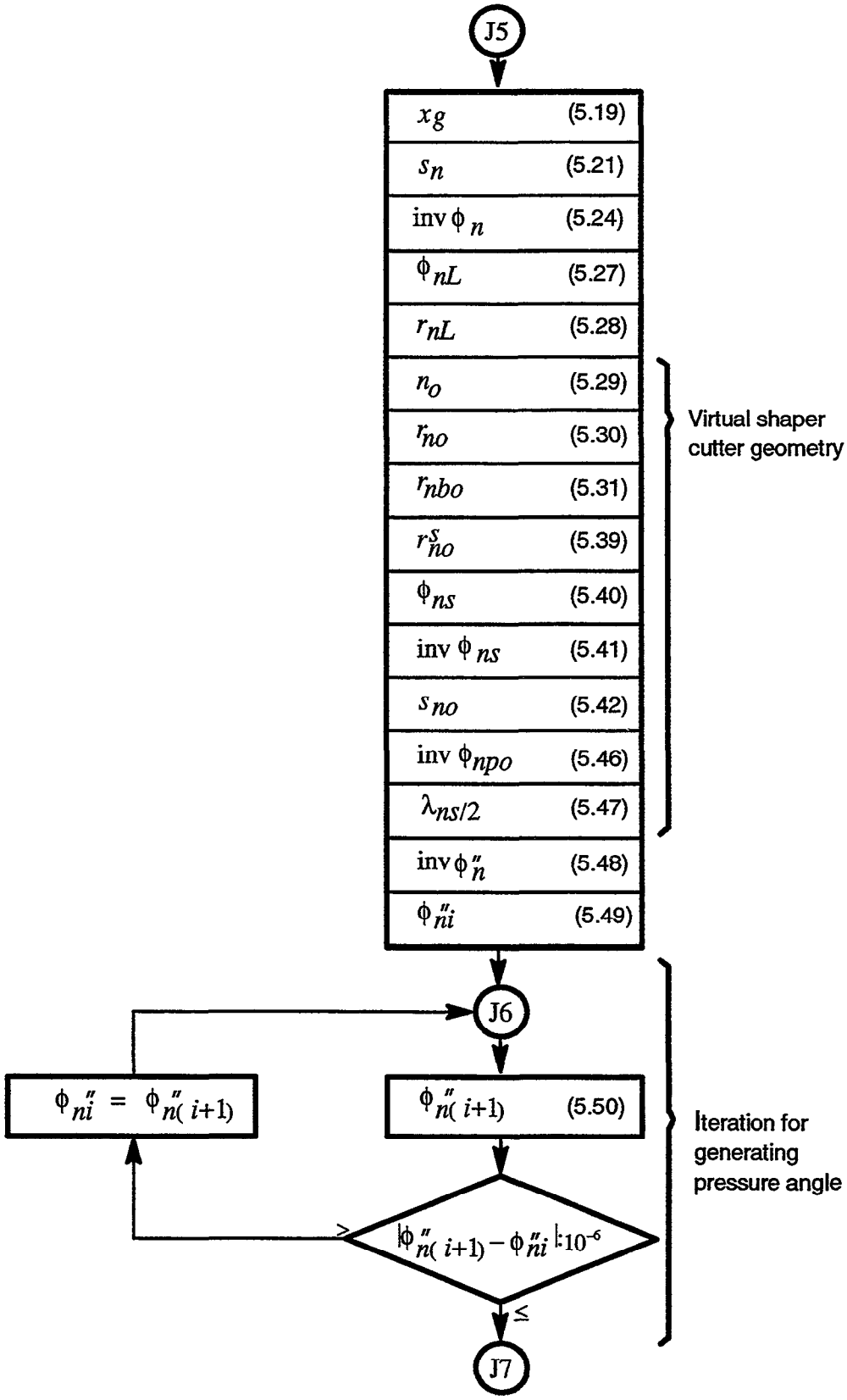


Figure 1 *(continued)*

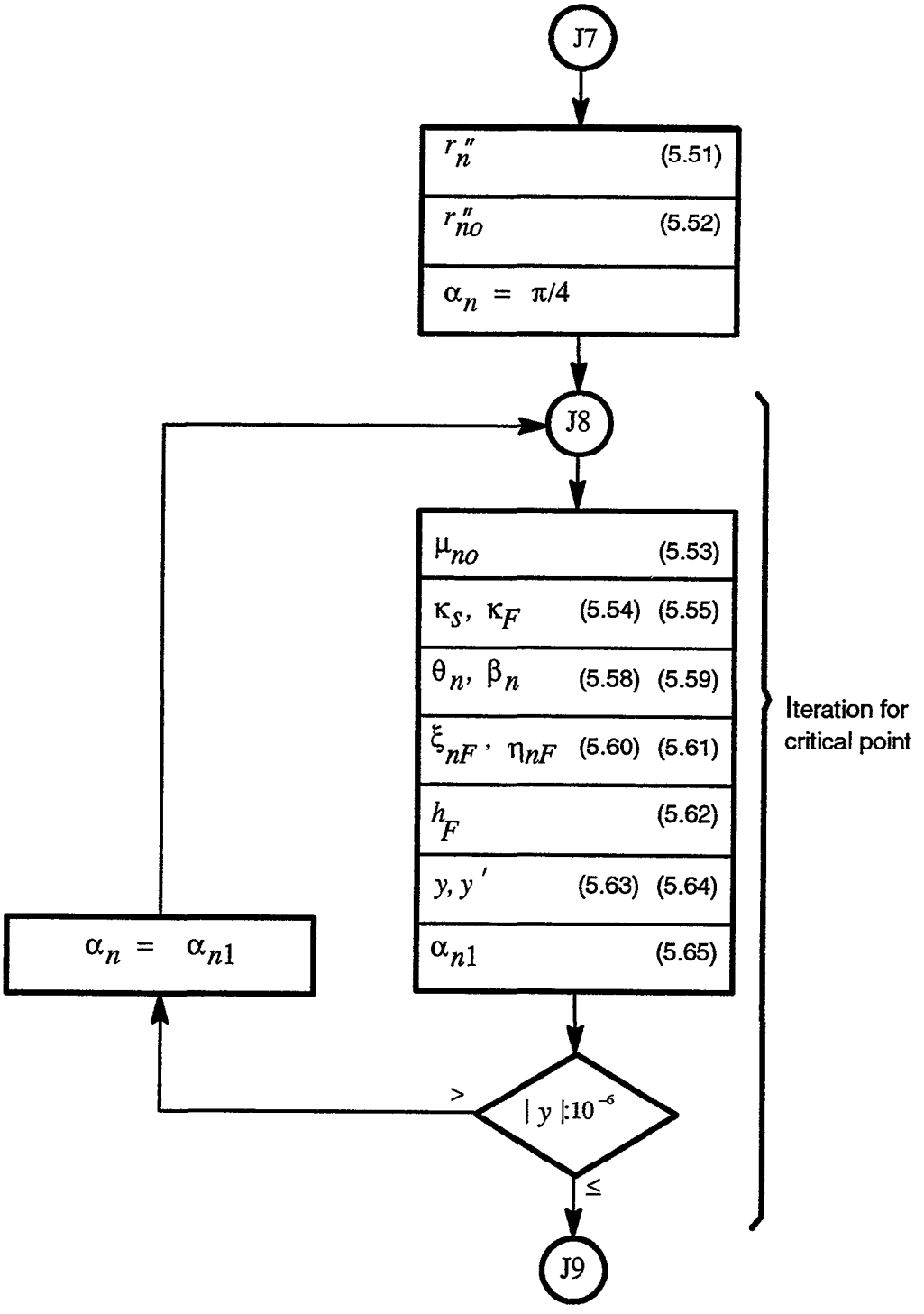


Figure 1 *(continued)*

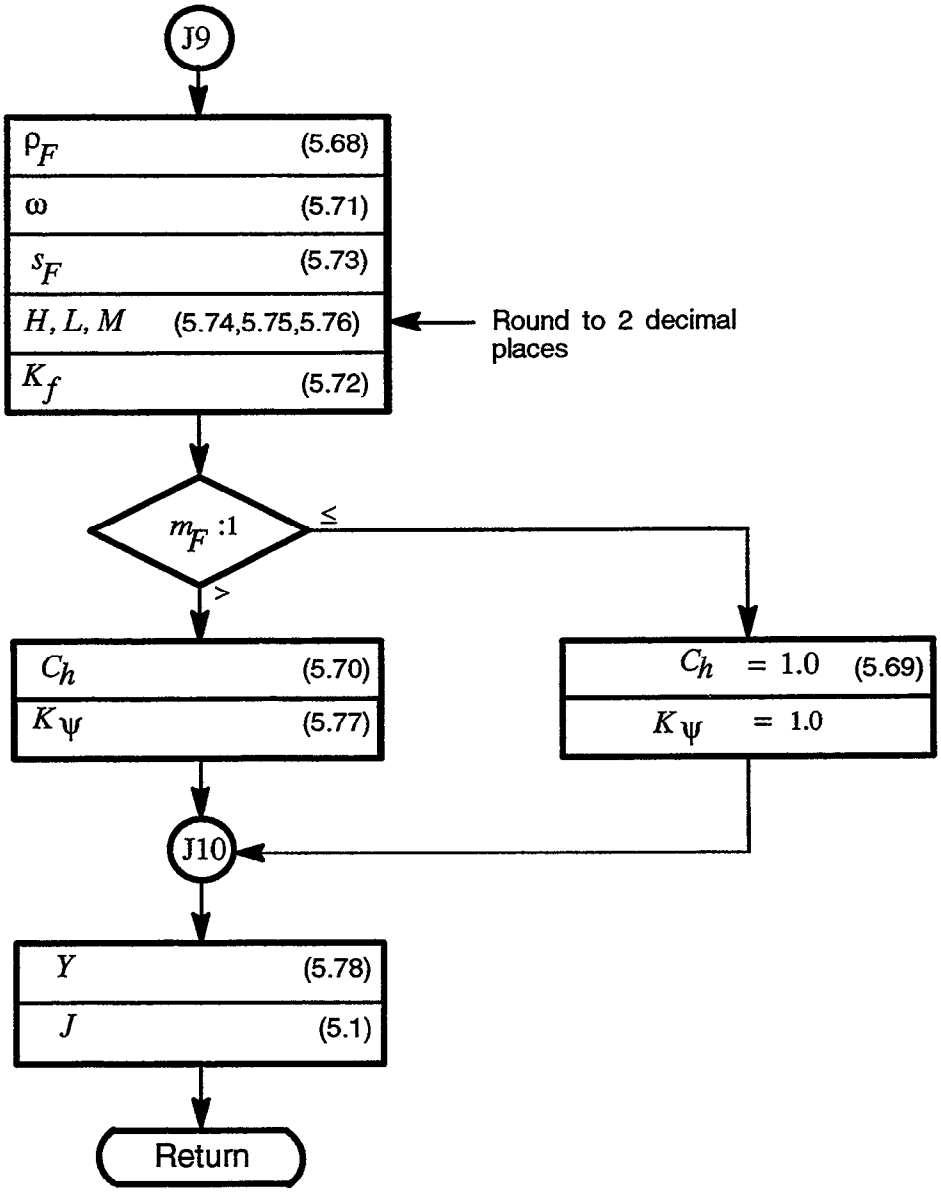


Figure 1 *(concluded)*

5 Cutting tool geometry

Figures 2 through 12 depict the geometry of the cutting tools used for the examples in this information sheet. These figures have dimensions taken from the cutting tools actually used to make the gears given in these examples. The procedure used to arrive at the dimensions given is beyond the scope of this information sheet.

5.1 Rack type cutting tools

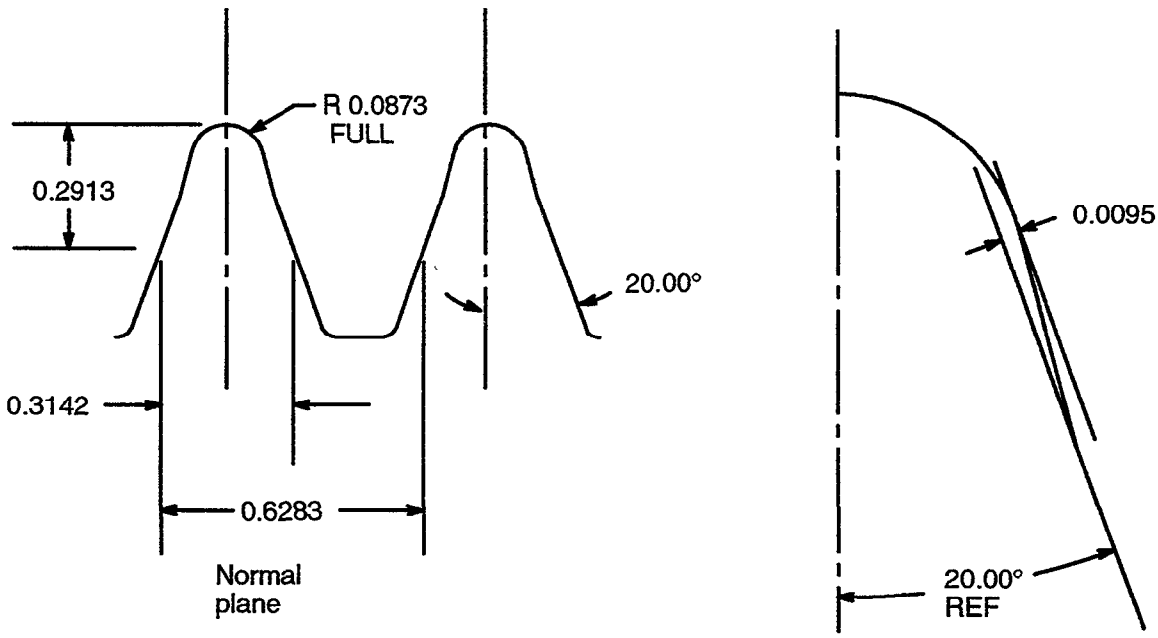
The hob drawings (figures 2, 3, 4, 6, 8, and 11) depict rack type cutting tools. The hob tooth is depicted in the normal plane with the hob addendum given to a circular tooth thickness of $\pi/(2\ P_{nd})$. All of these hobs are straight sided in the normal plane.

5.2 Pinion type cutting tools

The pinion type shaper cutter drawings (figures 5, 7,9, 10, and 12) depict pinion type cutting tools. The shaper cutter tooth is depicted in the normal plane with the shaper cutter addendum and chordal tooth thickness given at the pitch diameter of the tool. The dimensions apply when the tool is new and will change as it is sharpened back. All of these pinion type shaper cutters are involute in the transverse plane.

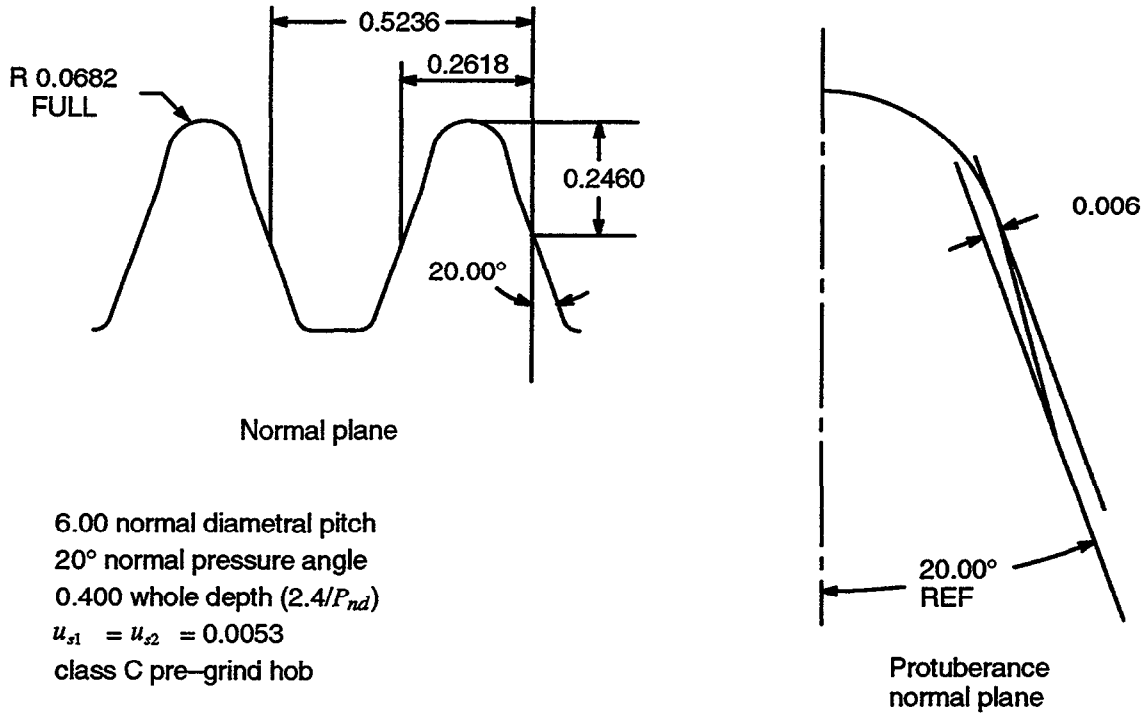
5.3 Cutting tool drawings

Figures 2 through 12 are simplified cutting tool drawings used in the examples, with dimensions in inches. These values have been normalized to 1.0 module for the input data in clause 3 (see figure 1). The stock allowance per side of tooth, u_s, for finishing, is given where applicable.



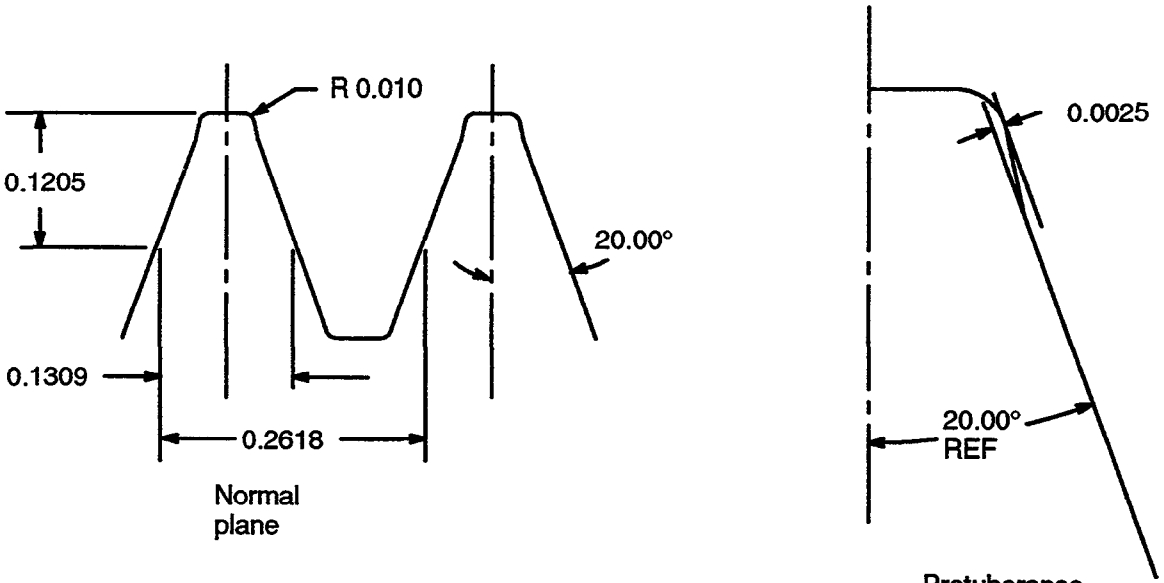
5.00 normal diametral pitch
20° normal pressure angle
0.472 whole depth ($2.36/P_{nd}$)
$u_{s1} = u_{s2} = 0.0080$
class C pre–grind hob

Figure 2 – Hob for example 3.1.1 and 3.1.2



6.00 normal diametral pitch
20° normal pressure angle
0.400 whole depth ($2.4/P_{nd}$)
$u_{s1} = u_{s2} = 0.0053$
class C pre–grind hob

Figure 3 – Hob for example 3.1.3 and 3.1.4



12.00 normal diametral pitch
20° normal pressure angle
0.200 whole depth ($2.4/P_{nd}$)
$u_{s1} = 0.0010$
ground pre–shave hob

Figure 4 – Hob for example 3.1.5

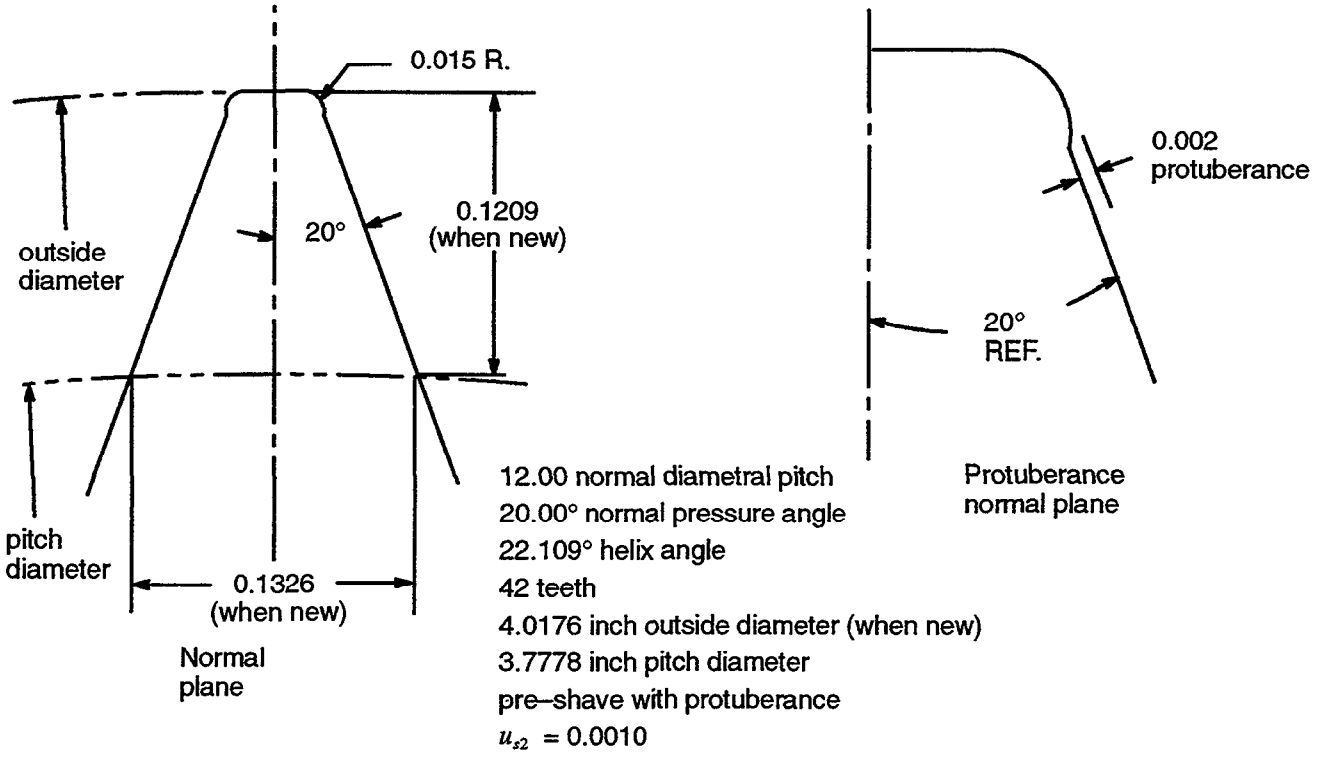


Figure 5 – Helical pinion type shaper cutter for example 3.1.5

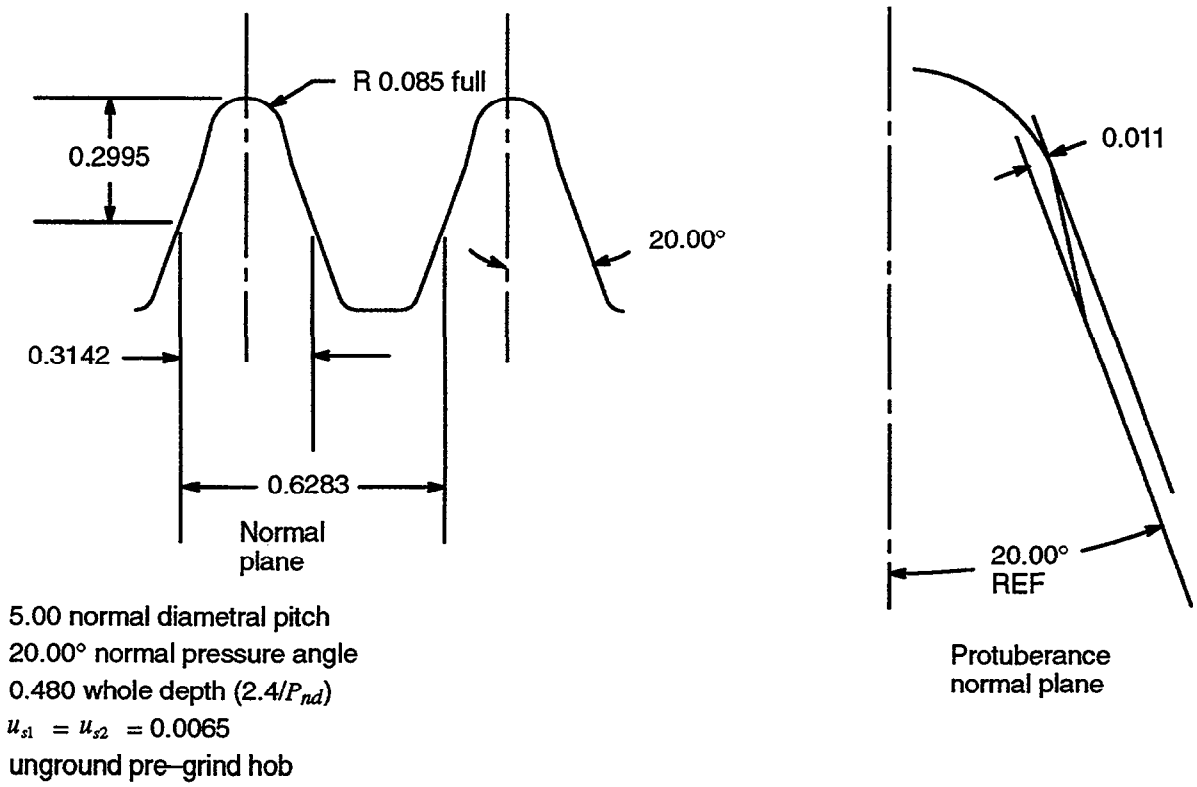
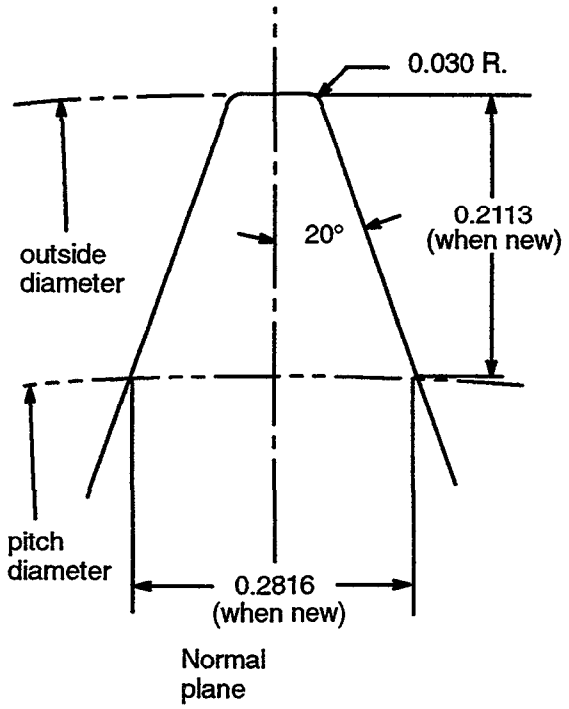
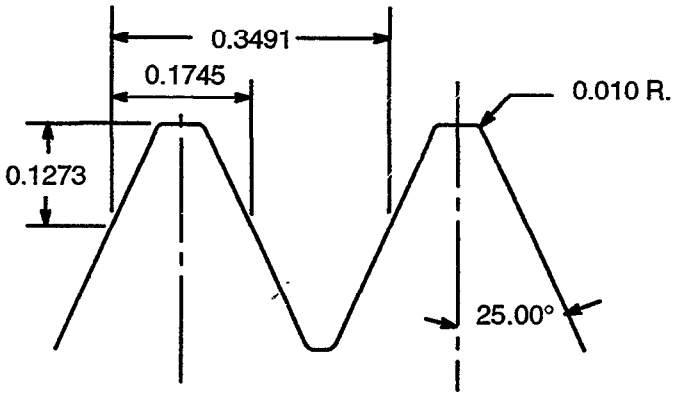


Figure 6 – Hob for examples 3.1.6 and 3.1.7



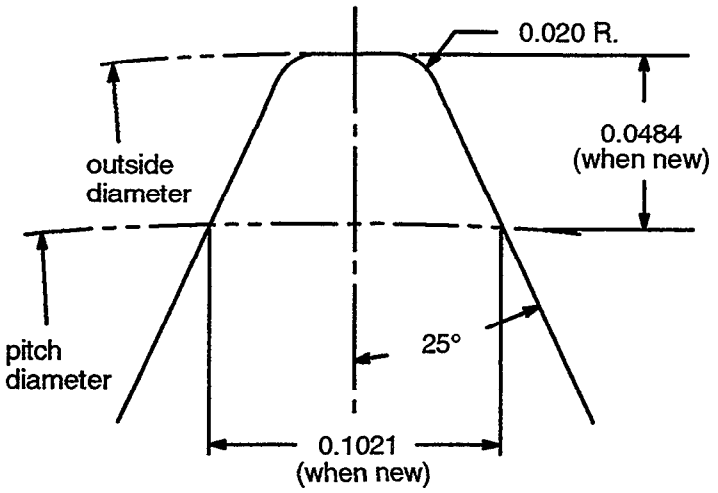
5.00 normal diametral pitch
20.00° normal pressure angle
0° helix angle
30 teeth
6.416 inch outside diameter (when new)
6.000 inch pitch diameter
finishing (no protuberance)

Figure 7 – Spur pinion type shaper cutter for example 3.1.7



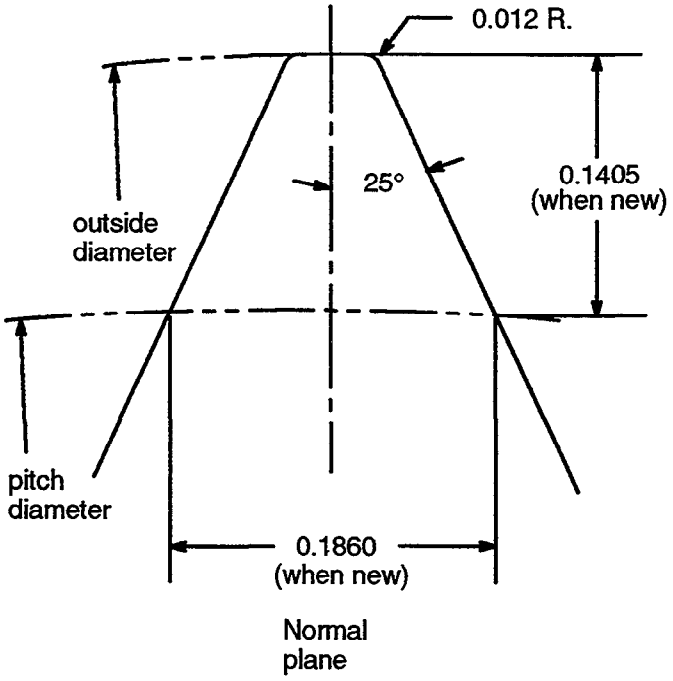
9.00 normal diametral pitch
25.00° normal pressure angle
0.233 whole depth ($2.10/P_{nd}$)
class A finish hob

Figure 8 – Hob for example 3.1.8



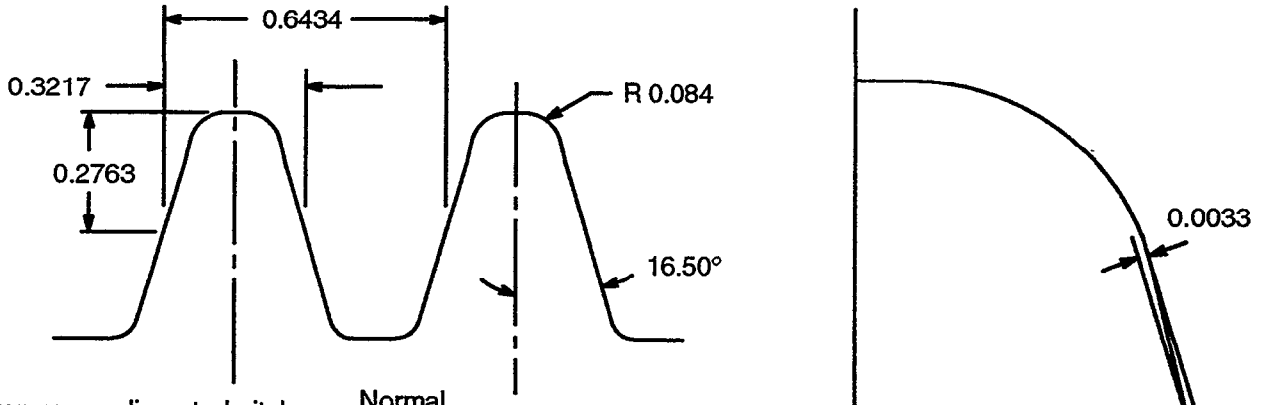
9.00 normal diametral pitch
25.00° normal pressure angle
17.7276° helix angle
36 teeth
4.295 inch outside diameter (when new)
4.1994 inch pitch diameter
finishing (no protuberance)

Figure 9 – Helical pinion type shaper cutter for examples 3.1.8 and 3.1.9



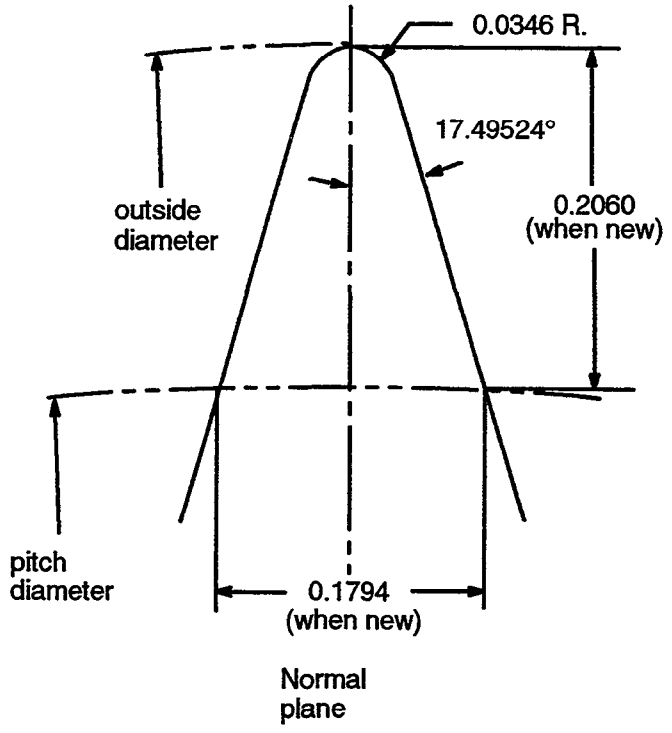
9.00 normal diametral pitch
25.00° normal pressure angle
17.7276° helix angle
36 teeth
4.4766 inch outside diameter (when new)
4.1994 inch pitch diameter
finishing (no protuberance)

Figure 10 – Helical pinion type shaper cutter for example 3.1.9



4.00 transverse diametral pitch
(corrected for 35° helix angle)
4.883098 normal diametral pitch
16.50° normal pressure angle
0.491 whole depth ($2.4/P_{nd}$)
class A pre–shave hob
u_{s1} = 0.0010
u_{s2} = 0.0003

Figure 11 – Hob for example 3.1.10



9.2376 normal diametral pitch
(8.00 transverse diametral pitch)
17.49524° normal pressure angle
(20.00° transverse pressure angle)
30° helix angle
32 teeth
4.409 inch outside diameter (when new)
4.000 inch pitch diameter
finishing (no protuberance)

Figure 12 – Helical pinion type shaper cutter for example 3.1.11



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